

The $SU(2)$ mid-plane on its small-field set: bounds and the missing comparison

After refereeing (GPT-6 Astra, 2026-09-28). Claim-by-claim review of Claude's derivation: **A1 ACCEPT** (with the inverse notation corrected); **A2 REFINE** (complete image pairing, explicit constants, endpoint range, and the distinction between a bridge and the interacting measure); **A3 REFINE** (local convexity survives with corrected midpoint curvature and defined derivative constants); **A4 REJECT** (Proposition 4 is false under Hypothesis I, already along a flat-holonomy path; its covariance, gauge comparison and Taylor estimates also have missing premises); **A5 REFINE** (the Peierls arithmetic survives with corrected curvature constant; the logarithmic size condition remains conditional). The accepted results are Lemmas 1–2 and the corrected local Lemma 3 below. The normalized small-field comparison remains open. The original proof and its attribution are preserved in git history.

Take one halving of direction 1 in $1 + 2$ dimensions, with gauge group $SU(2)$, heat time $t = \lambda_3 a = \hbar g_{\text{cl}}^2 a$, and

$$\varepsilon = t^{1/2-\delta}, \quad \eta = 2\varepsilon, \quad 0 < \delta < 1/6.$$

All operator norms below use the Euclidean norm on the three real colour components, $[T_a, T_b] = \epsilon_{abc} T_c$, $|T_a| = 1$. Distances on $SU(2)$ use the three-sphere of radius 2; Haar measure has total mass one. Inputs read at passage level: series/parallel, Proposition 1, Hypothesis P(α), Proposition 7; mid-plane order t , §§1 and 7, especially (16)–(18); exact midpoint, Theorem 1 and its image formula. This note concerns one finite mid-plane and bounds uniform in its area where explicitly stated.

1. Setting and the restriction

There are N sites, $2N$ edges and N faces on the periodic mid-plane. Use periods at least three to avoid repeated boundary incidences. In the mid-vertex gauge of Proposition 1 set $m_e = m_{*e} \exp \xi_e$, with m_{*e} the shortest-geodesic midpoint between P_e and Q_e^{-1} . Their distance is $|X_e| \leq \varepsilon$. Up to constants independent of ξ , the exponent is

$$S(\xi; U) = \frac{1}{t} \sum_e \left[d(P_e, m_e)^2 + d(m_e, Q_e^{-1})^2 - \frac{1}{2} |X_e|^2 \right] + \frac{1}{4t} \sum_g |\log m_{\partial g}|^2 + J(\xi; U). \quad (1)$$

Here J contains the full heat-kernel and Haar amplitudes. Its local ξ derivatives are bounded uniformly on a fixed chart away from the cut locus. Constants in J , including its bridge value at $\xi = 0$, can be transferred to the external normalization; only the bridge's squared-distance term necessarily vanishes there. Count each bridge denominator once, either in S or outside the integral. Include the old-face ratios in the external normalization $B(U)$.

Keep Claude's soft-barrier definition explicit:

$$W_\eta(\xi) = \sum_e w_\eta(|\xi_e|), \quad \mu_U(d\xi) = Z_s(U)^{-1} e^{-S(\xi; U) - W_\eta(\xi)} d\xi, \quad \mathcal{D}_s = -\log Z_s + B(U). \quad (2)$$

The domain is $\prod_e B(0, c_0)$; w_η is C^2 , convex and radial, vanishes for $r \leq \eta$, and diverges at c_0 . Choose its boundary decay so that the integrations by parts used below have vanishing boundary terms and finite Fisher information. It is fixed independently of U in these midpoint coordinates. Its support extends beyond $\Omega_\eta = \{\max_e |\xi_e| \leq \eta\}$. A hard restriction to Ω_η defines a different partition function. Both choices are convex restrictions, but their covariance lower bounds need different boundary arguments.

The interacting law includes every mid-face weight. Lemma 2 bounds the single-edge bridge before these weights are inserted. Transferring that tail bound to (2), bounding the probability that any of $2N$ edges is large, and comparing the two partition functions are additional tasks. Even for independent bridges the union bound carries a factor $2N$.

2. Lemma 1: covariant inverse decay (A1 ACCEPT)

Let C_A be the oriented face-edge incidence operator with each entry an adjoint rotation of the background connection. Set $H_A = 4I + \frac{1}{2}C_A^*C_A$. For every background,

$$4I \leq H_A \leq 8I. \quad (3)$$

If $K = H_A + E$, $E = E^*$ has range one in the graph of edges sharing a face, and $\|E\| \leq \kappa < 4$, then, writing $d = d(e, e')$,

$$\|(K^{-1})_{ee'}\| \leq \frac{1}{4-\kappa} \left(\frac{2+\kappa}{6} \right)^d. \quad (4)$$

In particular the original range $\kappa < 2$ is valid, and at $\kappa = 0$ the bound is $\frac{1}{4}3^{-d}$. Here K denotes the perturbed Hessian; K^{-1} denotes its inverse.

Proof. For each face, $|(C_A\xi)_g|^2 \leq 4 \sum_{e \in \partial g} |\xi_e|^2$. Each edge belongs to two faces, so $\|C_A\xi\|^2 \leq 8\|\xi\|^2$. Thus $\|K - 6I\| \leq 2 + \kappa < 6$. The n th power of $K - 6I$ has zero (e, e') block for $n < d$, and each remaining block has norm at most $(2 + \kappa)^n$. Sum the Neumann series about $6I$ to obtain (4). The count of faces per edge makes every constant independent of N . $\square$

This is a bound on the sum of operator products at a given order. It supplies no bound of 3^{-L} for each individual path with the same constant after absolute path counting.

3. Lemma 2: heat kernel and bridge tail (A2 REFINE)

For $0 < t \leq 1$ and $0 \leq \theta \leq 2\pi$,

$$k_t(\theta) \leq C_+ t^{-5/2} e^{-\theta^2/(2t)}; \quad k_t(\theta) \geq c_- t^{-3/2} e^{-\theta^2/(2t)} \quad (0 \leq \theta \leq 1). \quad (5)$$

Here are explicit, deliberately loose constants. Put $a_n = 4\pi n$, $b_n = (4n + 2)\pi$, and

$$\begin{aligned} A_0 &= \pi \left[1 + 2 \sum_{n \geq 1} (1 + (a_n + \pi)^2) e^{-(a_n^2 - 2\pi a_n)/2} \right], \\ A_1 &= 2\pi \sum_{n \geq 0} (1 + (b_n + \pi)^2) e^{-b_n(b_n - 2\pi)/2}, \\ C_+ &= e^{1/8} \sqrt{8\pi} \max(A_0, A_1), \quad c_- = \sqrt{8\pi}. \end{aligned} \quad (6)$$

Written image check. Write $f_t(u) = u e^{-u^2/(2t)}$. The exact image formula is

$$k_t(\theta) = e^{t/8} \sqrt{8\pi} t^{-3/2} \frac{\sum_{w \in \mathbb{Z}} f_t(\theta - 4\pi w)}{\sin(\theta/2)}.$$

On $[0, \pi]$, pair $w = \pm n$. Their sum is $f_t(a_n + \theta) - f_t(a_n - \theta)$, which vanishes at zero. Since $|f'_t(u)| \leq (1 + u^2/t) e^{-u^2/(2t)}$, its absolute value is at most

$$2\theta(1 + (a_n + \pi)^2/t) e^{-(a_n - \theta)^2/(2t)}.$$

Use $\theta/\sin(\theta/2) \leq \pi$ and $a_n^2 - 2a_n\theta \geq a_n^2 - 2\pi a_n$ to obtain the A_0/t bound on the image quotient after extracting $e^{-\theta^2/(2t)}$.

On $[\pi, 2\pi]$, set $\psi = 2\pi - \theta \in [0, \pi]$. Pair **all** images as $w = -n$ and $w = n + 1$, $n \geq 0$. Each pair is $f_t(b_n - \psi) - f_t(b_n + \psi)$ and has magnitude at most

$$2\psi(1 + (b_n + \pi)^2/t)e^{-(b_n - \psi)^2/(2t)}.$$

Divide by $\sin(\psi/2) \geq \psi/\pi$ and use $(b_n - \psi)^2 - (2\pi - \psi)^2 \geq b_n(b_n - 2\pi)$. This gives A_1/t , including the antipodal limit $\psi = 0$. In particular the leading pair is

$$e^{-(4\pi^2 + \psi^2)/(2t)}[(2\pi - \psi)e^{2\pi\psi/t} - (2\pi + \psi)e^{-2\pi\psi/t}],$$

as in Claude's calculation. Pairing the remaining images too removes the apparent singularity from division by $\sin(\psi/2)$.

For $\theta \leq 1$, the absolute remainder relative to $\theta e^{-\theta^2/(2t)}$ is bounded by

$$2 \sum_{n \geq 1} (1 + (a_n + 1)^2/t) e^{-(a_n^2 - 2a_n)/(2t)} < \frac{1}{2}.$$

For completeness, $3 < \pi < 4$ bounds this sum by $2 \sum_{n \geq 1} (1 + 289n^2) e^{-56n^2} < 1/2$; use $t^{-1} e^{-b/t} \leq e^{-b}$ for $b \geq 1$, $t \leq 1$, followed by $n^2 \leq 4^n$, $e^{56} > 2^{56}$ and a geometric series. Together with $\theta/\sin(\theta/2) \geq 2$ this proves the lower bound with (6), also at zero by continuity. $\square$

Bridge conclusion. Assume $|X| \leq 1$, $r \geq |X|/2$, and $0 < t \leq 1$. For the normalized bridge of duration t ,

$$\beta\{d(m, m_*) \geq r\} \leq \min\left(1, C_2 t^{-7/2} e^{-2r(r-|X|)/t}\right), \quad C_2 = 32C_+^2/c_- . \quad (7)$$

Indeed at distance $\rho \geq r$ both endpoint distances are at least $\rho - |X|/2 \geq 0$. The two upper bounds at time $t/2$ divided by the lower bound at time t give the density bound with exponent $-[2(\rho - |X|/2)^2 - |X|^2/2]/t = -2\rho(\rho - |X|)/t$. This exponent decreases with ρ for $\rho \geq |X|/2$; integrate against Haar measure of mass one. At $r = \eta = 2\varepsilon$ and $|X| \leq \varepsilon \leq 1$ the gain is $e^{-4\varepsilon^2/t}$. This proof uses the two endpoint triangle inequalities through m_* .

4. Lemma 3: a corrected local convexity theorem (A3 REFINE)

The midpoint background face logarithm $Y_g = \log(m_* \partial g)$ obeys $|Y_g| \leq 3\varepsilon$. To see this, compare its four edges to those of the transported lower layer. Each midpoint changes its edge by distance at most $\varepsilon/2$, and the lower-layer plaquette has distance at most ε . Bi-invariance and the product triangle inequality give $\varepsilon + 4\varepsilon/2 = 3\varepsilon$. The bound ε in the original sketch needs an extra premise. Individual background links can be large; transport them out of each local product before taking logarithms.

Here is a precise version of the local estimate, with constants defined by fixed-dimensional derivatives rather than unquantified BCH symbols. Fix $r_0 = 1/16$. On $|b|, |z| \leq r_0$ define

$$B(b, z) = d(e, e^{-b/2} e^z)^2 + d(e, e^{b/2} e^z)^2 - |b|^2/2,$$

and on $|y|, |z_i| \leq r_0$ define

$$F(y, z_1, \dots, z_4) = \frac{1}{4} |\log(e^y e^{z_1} \dots e^{z_4})|^2.$$

All these products lie in a fixed injectivity chart. Let L_B be the supremum of the derivative of $D_z^2 B$ with respect to (b, z) , using the sum norm on the input blocks and operator norm on the output. Let L_F be the analogous supremum of the derivative of $D_{(z_1, \dots, z_4)}^2 F$. These are finite constants on the indicated compact sets. At the origin these Hessians are $4I$ and the block matrix with every block $I/2$, respectively.

Let J_B, J_F be supremum norms of the edge and face Hessians of the corresponding local heat-kernel/Haar log amplitudes for $0 < t \leq 1/2$ on the same charts; omit terms independent of ξ . They are finite: the zero-image Jacobian factors are analytic on these compact sets, and

each differentiated nonzero image has a polynomial in $1/t$ times $e^{-c/t}$ with $c > 0$. The positive zero-image quotient bounds the logarithm's denominator away from zero for small t ; positivity and compactness handle the remaining closed time interval. Define

$$J_* = J_B + 2J_F, \quad C_3 = L_B + 8L_F + J_*. \quad (8)$$

These supremum definitions fix the constants independently of the plane; a numerical value is unnecessary for the following explicit condition. For $3\varepsilon \leq r_0$, $\rho = |\xi|_\infty \leq r_0$, $0 < t \leq \min(1/2, \varepsilon)$, the **unbarriered** action satisfies

$$\|t \nabla_\xi^2 S - H_A\| \leq L_B(\varepsilon + \rho) + 2L_F(3\varepsilon + 4\rho) + tJ_* \leq C_3(\varepsilon + \rho). \quad (9)$$

Here C_A uses the signed adjoint transports in the ordered background face products. At each face the four rotations are isometries. Applying the mean value theorem to the two Hessians just defined gives the local bounds; summing face quadratic forms costs a factor two because every edge belongs to two faces. This proves (9) directly and supplies the derivative control missing from the BCH sketch.

Choose $c_0 \leq r_0$ with $C_3(\varepsilon + c_0) < 4$, and $2\varepsilon < c_0$. Then S is uniformly convex on the product of balls of radius c_0 ; $S + W_\eta$ has the same lower Hessian bound. On Ω_η the unbarriered error is at most $3C_3\varepsilon$. The soft measure (2) has the weaker pointwise error $C_3(\varepsilon + c_0)$ on its full support. Its barrier Hessian can be arbitrarily large. These distinctions matter in §5.

5. Proposition 4: verdicts and the corrected conditional implication

The original Hypothesis I asked for a C^2 path $U(s)$ from 1 to U with $|X_p(s)| \leq 2\varepsilon$,

$$\sum_p |\dot{X}_p(s)|^2 \leq C_I \sum_p |X_p(1)|^2, \quad |\ddot{X}_p(s)| \leq C_I \sum_{p' \sim p} |X_{p'}(1)|^2. \quad (10)$$

A4 overall: REJECT, with a counterexample. On a fixed periodic plane choose identical adjacent flat layers with commuting holonomies $e^{s\theta_2 T_3}, e^{s\theta_3 T_3}$. Constant links $e^{s\theta_j T_3/N_j}$ realize the path. Every plaquette flux, flux velocity and flux acceleration is zero, so (10) holds. Nevertheless, the fixed-volume Laplace calculation in (16)–(17) of the mid-plane note gives

$$\mathcal{D}_s(U_\theta, t) - \mathcal{D}_s(1, t) = \Omega_N(\theta) + o_N(1), \quad \Omega_N(\theta) > 0 \quad \text{for a nontrivial adjoint holonomy.} \quad (11)$$

The same leading term holds for (2): the barrier equals zero in a ball of radius η , and $\eta/\sqrt{t} = 2t^{-\delta} \rightarrow \infty$; rescaling by $\sqrt{t}$ recovers the full Gaussian determinant at fixed N . Uniform convexity about this zero-flux saddle controls the discarded tail at fixed N . The proposed right-hand side $C\varepsilon \sum_p |X_p|^2/t$ is exactly zero. Hypothesis I therefore allows data contradicting the claimed theorem. A winding term or an additive volume remainder must be included, with its own estimate. The actual $P(\alpha)$ allows $Ct^\alpha \sum_p 1$ and a size clause; those features distinguish it from the rejected stronger assertion.

A4(a), linear term: ACCEPT in a smooth local chart at 1. Global conjugation rotates every Lie-algebra link tangent by the adjoint representation. Its invariant covectors vanish, so the first derivative of the gauge-invariant partition function at 1 is zero. The same argument works for independent flux coordinates if such a chart has been supplied. Plaquette logarithms alone omit torus holonomies and satisfy compatibility constraints; writing a global ∇_X requires additional coordinates. Gauge invariance permits the quadratic flat holonomy response (11).

A4(b), differentiation: ACCEPT with its domain hypotheses. For a fixed coordinate domain and a U -independent barrier, differentiating under the finite integral gives, for any boundary coordinates u_i ,

$$\partial_i \partial_j \mathcal{D}_s = E[S_{ij}] - \text{Cov}(S_i, S_j) + B_{ij}. \quad (12)$$

Uniform local domination justifies differentiation. A moving hard boundary needs boundary terms, unless it is first represented in a fixed domain. Equation (12) alone provides no local, independent flux coordinates with plane-uniform derivative bounds.

A4(b), covariance upper bound: REFINE. Put $V = S + W_\eta$ and let κ bound the unbarriered Hessian error on the **whole support**. The Brascamp–Lieb inequality gives

$$\text{Cov}_\mu(\xi) \leq E[(\nabla^2 V)^{-1}] \leq t(H_A - \kappa I)^{-1}. \quad (13)$$

Convex barriers preserve this upper bound. For a hard convex restriction one can obtain it by convex approximation. In (2) the available choice is $\kappa = C_3(\varepsilon + c_0)$; replacing it by $3C_3\varepsilon$ requires controlling the exterior region.

A4(b), covariance lower bound: REJECT as used. With the stated integration-by-parts hypotheses, the score identity and matrix Cauchy–Schwarz give

$$\text{Cov}_\mu(\xi) \geq (E[\nabla^2 V])^{-1}, \quad E[\nabla V \nabla V^T] = E[\nabla^2 V]. \quad (14)$$

The right-hand side contains $E[\nabla^2 W_\eta]$. Its sign increases the Fisher information and decreases this lower bound. Discarding it has the wrong direction for the asserted lower estimate. A tail probability alone bounds neither an unbounded barrier Hessian nor the Fisher information. Hard truncation produces boundary terms; already a one-dimensional centred Gaussian truncated to a finite symmetric interval has variance strictly smaller than its untruncated variance. To recover a two-sided comparison one needs, for example,

$$\|tE[\nabla^2 S] - H_A\| \leq \kappa, \quad t\|E[\nabla^2 W_\eta]\| \leq b, \quad (15)$$

in addition to the pointwise lower bound in (13). These give the conditional estimate

$$\|\text{Cov}_\mu(\xi) - tH_A^{-1}\| \leq \max \left\{ \frac{t\kappa}{4(4 - \kappa)}, \frac{t(\kappa + b)}{4(4 + \kappa + b)} \right\}. \quad (16)$$

It follows by inversion of the Loewner bounds and $H_A \geq 4I$. The original proof established neither (15) with $b = O(\varepsilon)$ nor the required interacting tail estimate. The displayed Brascamp–Lieb inequality is the standard input (Brascamp–Lieb 1976, metadata); (14)–(16) follow by the written score calculation above.

A4(b), covariant-to-abelian comparison: REJECT as justified. The kernel $M_A H_A^{-1} M_A^T$ carries adjoint transport between distinct face frames. Gauge transformations act independently at these frames; comparison with a fixed abelian matrix needs specified identifications. Only traced closed products are gauge invariant by themselves. Even for a closed product, curvature bounds control contractible loops; a winding loop can have nontrivial holonomy with every $X_p = 0$. Finally (4) controls the total operator power, whereas taking absolute values path by path loses its cancellations and introduces path multiplicity. The proposed $\sum_L L^2 3^{-L}$ argument omitted all three issues. Needed: a gauge-covariant bulk kernel comparison with summable spatial moments, and a separate winding remainder.

A4(b), nonlinear score remainder: REFINE. If local derivatives actually satisfy $E|\nabla_\xi \langle v, R \rangle|^2 \leq C(\varepsilon + \eta)^2 \|v\|^2 / t^2$ on a support with curvature error κ , then (13) implies $\text{Var} \langle v, R \rangle \leq C(\varepsilon + \eta)^2 \|v\|^2 / [t(4 - \kappa)]$. Cauchy–Schwarz bounds the cross covariance. The Taylor bound on R requires its derivative version and a controlled boundary chart; its value bound alone supplies neither.

A4(c), Taylor step: REFINE. A precise surviving implication is as follows. Suppose that winding has been separated, a genuine C^2 coordinate path $X(s)$ satisfies (10), and $h = \mathcal{D}_s - \mathcal{D}_0 - W_N$ is a function of those coordinates. Assume $X(0) = 0$, $\nabla h(0) = 0$, and along the path

$$\|\nabla^2 h\| \leq A\varepsilon/t, \quad \sup_p |\nabla_p h| \leq B\varepsilon/t. \quad (17)$$

Let q_I bound the number of p for which a fixed p' appears in $p' \sim p$. The exact chain rule and Taylor's integral formula give

$$|h(X(1)) - h(0)| \leq \frac{C_I}{2}(A + Bq_I)\frac{\varepsilon}{t} \sum_p |X_p(1)|^2. \quad (18)$$

Indeed the Hessian term integrates to at most $AC_I\varepsilon\|X(1)\|^2/(2t)$ and the gradient-acceleration term to at most $BC_Iq_I\varepsilon\|X(1)\|^2/(2t)$. An operator norm estimate alone gives an ℓ^2 gradient bound; it does not imply the per-plaquette bound in (17). A uniformly summable kernel estimate would supply that missing norm. For abelian data the linear flux path is available only in a compatible coordinate sector with the harmonic data fixed. All constants in (18) are explicit in the supplied hypotheses; proving them uniformly in the plane is still part of the problem.

6. What stays open for the weak (18) of the mid-plane note (A5 REFINE)

- (a) **Small-field comparison.** The tasks include barrier moment control, a background-covariant local response estimate, independent admissible coordinates or a replacement for the global path, and winding control. The original list of only Hypothesis I and large fields was incomplete. The formal order- t coefficients retain their status as formal coefficients; this review supplies no normalized remainder for them.
- (b) **Large fields: the failed naive Peierls arithmetic is accepted.** Using the extra premise $|Y_g| \leq \varepsilon$, the face loss at $|\xi_e| \leq 2\varepsilon$ is $\exp[-(\varepsilon + 8\varepsilon)^2/(4t)] = e^{-81\varepsilon^2/(4t)}$ per affected face, against bridge gain $e^{-4\varepsilon^2/t}$ per large edge. Since an edge touches two faces, the crude worst-case loss is $81\varepsilon^2/(2t)$ per edge and the net exponent is $+(73/2)\varepsilon^2/t$. The comparison grows rather than decays. With only the coarse small-field assumption, §4 gives $|Y_g| \leq 3\varepsilon$; replace 81 by 121 and $73/2$ by $113/2$. These are bounds on the exponential part; heat-kernel prefactors add powers of t , which cannot repair the wrong sign as $\varepsilon^2/t = t^{-2\delta} \rightarrow \infty$. Shared faces can reduce the loss for particular sets; the crude bound suffices to expose failure of the proposed uniform argument. A normalized connected-set or other large-field estimate is still required. The soft barrier also needs comparison with a genuine small-field partition function.
- (c) **Logarithmic size: conditional, with the rate corrected.** For flat Gaussian backgrounds the explicit walk count (17) in the mid-plane note proves $0 \leq \Omega_N \leq (3N/n_*)3^{-n_*}$, $n_* = \min(N_2, N_3)$. More generally, if a winding estimate

$$|W_N| \leq Ce^{-\gamma n_*} \sum_p (1 + |X_p|^2/t) \quad (19)$$

has been proved with $\gamma > 0$ independent of N, t , then $n_* \geq (\alpha/\gamma)\log(1/t) + C_0$ suffices for $P(\alpha)$, with the constant multiplied by $e^{-\gamma C_0}$. A decay factor $q = (2 + \kappa)/6$ would give $\gamma = \log(6/(2 + \kappa))$; $\log 3$ is the value at $\kappa = 0$. With fixed positive κ one must use the smaller rate. If $\kappa = O(\varepsilon)$, the $\log 3$ leading coefficient can be recovered with a bounded correction because $\varepsilon \log(1/t) \rightarrow 0$. Polynomial prefactors in n_* need absorption into a slightly smaller rate or a $\log \log(1/t)$ correction. Lemma 1 alone proves (4); a winding estimate for the normalized interacting free energy still requires a separate argument.

7. Part B: quasi-local response and the obstruction to plaquette telescoping

Result, GPT-6 Astra, 2026-09-28. A covariance version of Lemma 1 gives a plane-uniform locality theorem for local perturbations of the potential, under the explicit analytic hypotheses below. Converting this to independent plaquette perturbations needs a further gauge estimate. The natural uniformly bounded local right inverse of curl already fails at the abelian linearization, as (26) proves. Thus this step leaves Hypothesis I unreplaced; it isolates the cancellation a successful replacement must establish.

7.1 Source check and scope

Helffer–Sjöstrand (1994), *On the correlation for Kac-like models in the convex case*, J. Stat. Phys. **74**, 349–409, treats dimension-dependent Gibbs measures that are controlled perturbations of harmonic potentials (reading label: **publisher metadata and abstract**; the full journal text was behind subscription access in this check). Helffer’s 1993 author report explicitly presents their joint work (reading label: **passage**, §4, Theorem 4.2, and §5, printed pp. XII-13–17). It states exponential correlation decay for sufficiently small coupling in a nearest-neighbour harmonic model with a C^3 interaction having bounded derivatives through order three. Section 5 explains the elliptic vector equation and the additional weighted estimates needed for spatial decay. This checks a concrete source scope; the general finite-range statement needed here is proved below with its own hypotheses. The author report also addresses thermodynamic limits. No source passage checked here supplies gauge charts, barrier comparison or plaquette telescoping. Bibliography entries: `HelfferSjostrand1994`, `Helffer1993Kac`.

7.2 A covariance theorem with the same decay constants

After refereeing Theorem 5 (Claude, 2026-09-28). ACCEPT. The representation (22) is the Helffer–Sjöstrand formula; $\mathcal{B} \geq 6I$ holds because $L \geq 0$ and the on-site Hessians are nonnegative, and $\mathcal{B}^{-1}$ keeps the edge index; $\mathcal{R}$ is multiplication by a range-one matrix function of norm at most $2 + \kappa$; the Neumann tail gives $\frac{1}{6} \sum_{n \geq d} q^n = q^d/(4 - \kappa)$, and Cauchy–Schwarz in $L^2(\mu)$ gives (21). The application to the barrier measure is correctly left as a stated approximation condition.

Theorem 5 (local-potential response). Let E be the finite edge graph and $\mu \propto e^{-V} d\xi$ on $\mathbb{R}^{3E}$, with $V = S + \sum_e w_e(\xi_e)$ smooth and confining, w_e convex. Assume the usual closed weighted-gradient realization, so integration by parts and the differentiated Poisson equation hold; smooth potentials with bounded Hessians and a uniform positive lower bound suffice. Allow limits of such measures only when covariances and the gradient norms appearing below converge. Suppose, everywhere on the domain,

$$\|t\nabla^2 S - 6I\| \leq 2 + \kappa, \quad 0 \leq \kappa < 4, \quad (\nabla^2 S)_{ee'} = 0 \quad \text{if } d(e, e') > 1. \quad (20)$$

For smooth f, g with gradients supported in edge sets R, T , define $a_f = \|\nabla f\|_{L^2(\mu)}$ and $a_g = \|\nabla g\|_{L^2(\mu)}$. Then, uniformly in the graph size,

$$|\text{Cov}_\mu(f, g)| \leq \frac{t}{4 - \kappa} q^{d(R, T)} a_f a_g, \quad q = \frac{2 + \kappa}{6} < 1. \quad (21)$$

The on-site convex Hessians can be large; the bound uses their sign, without treating them as a small perturbation.

Proof. Write $L = -\Delta + \nabla V \cdot \nabla$, nonnegative in $L^2(\mu)$. Uniform convexity gives a gap at least $(4 - \kappa)/t$ for this auxiliary diffusion. Solve $Lu = g - Eg$ on the orthogonal complement of constants. Differentiating gives

$$(LI + \nabla^2 V)\nabla u = \nabla g, \quad \text{Cov}(f, g) = \langle \nabla f, (LI + \nabla^2 V)^{-1} \nabla g \rangle_{L^2(\mu)}. \quad (22)$$

The first equality is the commutator identity $\nabla Lu = (LI + \nabla^2 V)\nabla u$; the second follows by integration by parts in $\langle f - Ef, Lu \rangle$. This is the covariance representation motivating the Helffer–Sjöstrand method.

On the vector Hilbert space set

$$\mathcal{B} = tLI + 6I + t \operatorname{diag}_e \nabla^2 w_e, \quad \mathcal{R} = t\nabla^2 S - 6I.$$

The operator $\mathcal{B}$ is block diagonal in the edge **index**, although each scalar diffusion acts on all configuration variables. It satisfies $\mathcal{B} \geq 6I$; $\mathcal{R}$ has range one and norm at most $2 + \kappa$. Therefore

$$t(LI + \nabla^2 V) = \mathcal{B} + \mathcal{R}, \quad (\mathcal{B} + \mathcal{R})^{-1} = \sum_{n \geq 0} (-\mathcal{B}^{-1}\mathcal{R})^n \mathcal{B}^{-1}.$$

Its block from T to R has zero contributions for $n < d(R, T)$. The remaining operator norm is bounded by $\frac{1}{6} \sum_{n \geq d} q^n = q^d / (4 - \kappa)$. Insert this in (22), with the factor t from the inverse scaling, to prove (21). Smooth approximation and convergence extend the inequality to the stated limits. $\square$

For the local action in §4, (20) holds on its chart with $\kappa = C_3(\varepsilon + c_0)$, independently of the background connection. Application to the singular soft-barrier measure (2) additionally needs a realization or approximation satisfying the analytic hypotheses of Theorem 5 with the same finite-range bounds. Lemma 3 is a chart estimate; constructing that approximation remains an explicit condition here. This avoids importing a whole-space theorem into a bounded chart without specifying its boundary domain. Taking $\kappa \leq 1$ gives the concrete constants $t/3$ and decay 2^{-d} as looser bounds in (21). These rates concern mid-plane fluctuations and their auxiliary diffusion.

7.3 What locality of a free-energy increment means

Suppose $V_{u,v} = V_0 + uh_R + vk_T$, $(u, v) \in [0, 1]^2$, satisfies Theorem 5 uniformly with the same t, κ . Here the gradients of h_R, k_T have supports R, T , and let $a = \sup_{u,v} \|\nabla h_R\|_{L^2(\mu_{u,v})}$, $b = \sup_{u,v} \|\nabla k_T\|_{L^2(\mu_{u,v})}$. For $F(u, v) = -\log \int e^{-V_{u,v}}$, differentiating twice yields $F_{uv} = -\operatorname{Cov}(h_R, k_T)$. Consequently

$$|F(1, 1) - F(0, 1) - F(1, 0) + F(0, 0)| \leq \frac{tab}{4 - \kappa} q^{d(R, T)}. \quad (23)$$

Thus the increment caused in R depends exponentially weakly on a remote change in T . For a local observable f_T independent of u , the same argument gives

$$|E_1 f_T - E_0 f_T| \leq \frac{t}{4 - \kappa} q^{d(R, T)} \sup_u \|\nabla f_T\|_{L^2(\mu_u)} \sup_u \|\nabla h_R\|_{L^2(\mu_u)}. \quad (24)$$

These statements also hold for nonlinear parameter paths if the mixed potential derivative vanishes for disjoint supports, with h_R, k_T replaced by the parameter scores. Local external normalizations have zero mixed derivative for disjoint supports. In a fixed local link chart, changing coarse links in a region changes only incident bridge and face potentials, after enlarging the region by a fixed number of cells. Subject to the analytic condition after Theorem 5, (23) is therefore the precise localized-increment statement for $\mathcal{D}_s$.

Tails are summable uniformly on these two-dimensional edge graphs. A crude bound on the number of edges at graph distance ℓ from one edge is $2(4\ell + 3)^2$. Thus

$$\sum_{\ell \geq r} 2(4\ell + 3)^2 q^\ell \leq C(q) q^{r/2}, \quad C(q) = 2 \sum_{\ell \geq 0} (4\ell + 3)^2 q^{\ell/2} < \infty. \quad (25)$$

Summing (23) over remote local scores with bounded gradient norms gives this tail, with their support-size and support-radius constants made explicit in a, b and the distance. The

unlocalized increment itself may have size proportional to the changed region; (23) controls its dependence on the distant environment.

7.4 The estimate that fails in the proposed telescoping

Plaquette variables obey compatibility relations. A single plaquette update at fixed surrounding data generally fails those relations. Even in the linearized embedded $U(1)$ sector on an $n \times n$ torus, with harmonic data fixed, infinitesimal link changes a and flux changes x satisfy $Ca = x$ and $\sum_p x_p = 0$. At momentum $k = (2\pi/n, 0)$ the curl symbol of the mid-plane note is $c(k) = (0, e^{2\pi i/n} - 1)$. For a flux mode supported at this momentum, every solution satisfies

$$\|a\|_2 \geq \frac{\|x\|_2}{2 \sin(\pi/n)} \geq \frac{n}{2\pi} \|x\|_2. \quad (26)$$

Real sine/cosine modes give the same bound. Hence a right inverse $a = \mathcal{R}x$ on admissible zero-mean fluxes cannot have norm bounded independently of n . An exponentially local kernel $\|\mathcal{R}_{ep}\| \leq Ce^{-\gamma d(e,p)}$ with C, γ independent of n would have bounded row and column sums by (25), and bounded ℓ^2 norm by the Schur test, contradicting (26). Changing two separated opposite plaquette fluxes can instead use a long string of changed links or a spatially spread representative. The supports and score norms in (23) then track that representative.

The precise missing step is a **curvature response estimate after gauge cancellations**, rather than a local right inverse. For example, in specified admissible flux coordinates with harmonic data retained and winding subtracted, one would need the normalized remainder $h = \mathcal{D}_s - \mathcal{D}_0 - W_N$ to have a Hessian kernel K_{pq} satisfying

$$\sup_p \sum_q e^{\gamma d(p,q)} \|K_{pq}\| \leq C\varepsilon/t, \quad (27)$$

together with compatible paths or local replacements whose total quadratic cost is bounded by $C \sum_p |X_p|^2$, and the required zero-flux/winding remainder. Equation (21) controls the link-score covariance. The cancellation of the large inverse-curl factors in the combined expression $E[S_{ij}] - \text{Cov}(S_i, S_j) + B_{ij}$ remains to be proved. The Gaussian case exhibits such cancellations; (26) therefore diagnoses failure of the naive coordinate conversion, without ruling out (27).

With those extra estimates, (18), or a compatible telescoping version of it, and (19) would give the weak $P(\alpha)$ bound on the selected large-plane regime. Theorem 5 and locality alone supply neither the $O(\varepsilon)$ comparison to the Gaussian response nor the winding bound, and Part A's barrier and large-field issues remain. This is the precise obstruction reached in Part B. All claims of $P(\alpha)$ and the normalized expansion (18) of the mid-plane note remain conditional.

8. Round 9: cancellation in flux response

After refereeing (Claude, 2026-09-28). §8.1 ACCEPT, checked in full: Woodbury gives $G = \frac{1}{4}(I - C^*PC)$, hence (28); the slab form (29) vanishes at constant flux and gives $-|x|^2/(4t)$ for opposite layers, both as in Proposition 2; $8I + L = 12I - A$ and the walk series give the weighted row sum $1/(12 - 4e^\gamma) = \frac{1}{6}$ at $\gamma = \log(3/2)$, so (30) reads $\frac{1}{t}(\frac{5}{12} + \frac{1}{4}) = \frac{2}{3t}$. §8.2 ACCEPT as a formal statement: the equal-layer symbol $M(p) = -\lambda/(2(8 + \lambda))$ follows from (29), a flat neutral connection shifts it to $M(p + H)$, and the Coulomb lift $X^3(k)/(e^{i\kappa} - 1)$ of a soft flux produces the $1/\kappa$ residue in (32). The pole is a property of the fixed-frame chart (parallel transport of charged curvature), consistent with the gauge-invariant expectation of §6; it fixes that (27) must be posed covariantly. §8.3 ACCEPT.

8.1 Exact Gaussian reference (GPT-6 Astra, 2026-09-28)

Use the unrestricted abelian Gaussian in one slab with boundary links a_0, a_1 . Retain their harmonic components h_0, h_1 and impose the linear compatibility relations on all cut and transverse

fluxes. Put $x = Ca_0$, $y = Ca_1$, so $\sum x = \sum y = 0$. Independent zero-mean Fourier components and harmonic data give actual coordinates. The spatial kernels below represent bilinear forms on this constrained space, by restriction of an ambient kernel; eliminating one face or inserting a nonlocal orthogonal projector would change its row sums. The three colour copies in the linearization are identical. Set $L = CC^*$, $P = (8I + L)^{-1}$ and $G = H_0^{-1}$. The mean bridge link is $b = (a_0 + a_1)/2$ up to a gradient from direction-1 links. In uncentred mid-links m the normalized exponent and external old-face term are

$$S_0 = \frac{2}{t}\|m - b\|^2 + \frac{1}{4t}\|Cm\|^2, \quad B_0 = -\frac{\|x\|^2 + \|y\|^2}{8t}.$$

Assign half of each old-face ratio to each slab. Since $\text{Cov}_0(m) = tG$, differentiation in b gives

$$E_0[S_{0,bb}] - \text{Cov}_0(S_{0,b}, S_{0,b}) = \frac{1}{t}(4I - 16G) = \frac{4}{t}C^*PC, \quad G = \frac{1}{4}(I - C^*PC). \quad (28)$$

For any lift $\mathcal{R}$ with $C\mathcal{R} = I$ on admissible fluxes, $(C\mathcal{R})^*P(C\mathcal{R}) = P$ as a restricted bilinear form. The large factors in (26) disappear. Gaussian integration consequently gives

$$\begin{aligned} \mathcal{D}_{0,\text{slab}} &= \frac{1}{2t}\langle x + y, P(x + y) \rangle - \frac{\|x\|^2 + \|y\|^2}{8t} + \text{const}, \\ K_{\sigma p, \tau q}^0 &= \frac{1}{t} \left(P_{pq} - \frac{1}{4}\delta_{\sigma\tau}\delta_{pq} \right) I_3, \quad \sigma, \tau \in \{0, 1\}. \end{aligned} \quad (29)$$

Cut-flux directions at fixed x, y and all harmonic directions have zero response. Summing slabs reproduces Proposition 2 of the series/parallel note (passage). On the square torus $8I + L = 12I - A$, where A is nearest-neighbour adjacency. Its positive walk series yields $\sup_p \sum_q e^{\gamma d(p,q)} P_{pq} \leq (12 - 4e^\gamma)^{-1}$. For $\gamma = \log(3/2)$ and the pair distance $d((\sigma, p), (\tau, q)) = d(p, q) + |\sigma - \tau|$, (29) therefore gives

$$\sup_{\sigma, p} \sum_{\tau, q} e^{\gamma d} \|K_{\sigma p, \tau q}^0\| \leq \frac{1}{t} \left(\frac{1 + e^\gamma}{12 - 4e^\gamma} + \frac{1}{4} \right) = \frac{2}{3t}. \quad (30)$$

Assembly gives $4/(3t)$, independently of the periods. The Gaussian remainder h has zero kernel, hence satisfies (27) with $C = 0$; (30) bounds the reference response itself.

Mechanism. The Gaussian score identity produces the Schur complement (28), equivalently the Woodbury identity for $4I + C^*C/2$. Its two exterior curls annihilate gradients and harmonic shifts and turn every flux lift into the identity. Combining the direct Hessian and covariance before taking absolute values leaves a massive face resolvent. Normalized bridge denominators remove the cut-face response; the old-face normalization adds only the diagonal term in (29). This calculation uses the full Gaussian domain; a barrier requires its own correction.

8.2 Formal non-abelian response: a surviving soft transport term

Specify a chart before testing (27). Use face logarithms in fixed site frames, with the formal Coulomb lift $a = \mathcal{R}X + H + O(X^2)$, $\mathcal{R} = C^*L^{-1}$ on zero-mean modes; retain H and all nonlinear compatibility relations. This is a finite-volume formal chart about zero. Its cubic response already contains an inverse-curl pole. The following test concerns equal adjacent layers, zero cut flux, and $H = 0$.

Use the unrestricted zero-image expansion; barrier and image errors remain outside it. Write the normalized action as $S_0 + V_3 + V_4 + J + \dots$, with $V_j = \mathcal{S}_j/t$, and expand the BCH logarithm as $Z_1 + Z_2 + Z_3 + \dots$. Each squared mid-face contributes $\langle Z_1, Z_2 \rangle/(2t)$ and $(|Z_2|^2 + 2\langle Z_1, Z_3 \rangle)/(4t)$ to V_3, V_4 respectively. The bridge curvature contributes $-\sum_e |[X_e, \xi_e]|^2/(24t)$ to V_4 ; J includes the normalized Haar and heat-kernel amplitudes. With E_0 at the shifted Gaussian saddle, the first insertions and the required connected cubic pair are

$$\Delta\mathcal{D} = E_0V_3 + E_0V_4 - \frac{1}{2}\text{Cov}_0(V_3, V_3) + E_0J + \dots \quad (31)$$

One vertex is first order in an insertion parameter; under $[\ , \] \mapsto g[\ , \]$, the cubic is order g , while quartic and the connected pair are both order g^2 . Gaussian Wick contraction gives the classical terms $F_3/t, F_4/t$ in (15) of the order- t note (passage), including $-\langle\partial_\xi\mathcal{S}_3, G\partial_\xi\mathcal{S}_3\rangle/(2t)$. Thus a one-vertex quartic truncation alone would miss a necessary term. The complete classical cubic can be tested without listing its vertices. For equal layers, (29) has flux Hessian $M(p)/t$, where $M(p) = -\lambda(p)/(2(8 + \lambda(p)))$ and $\lambda(p) = 4 - 2\cos p_2 - 2\cos p_3$. At a constant flat neutral connection $a_j = H_jT_3$, the charged components have the exact classical quadratic symbol $M(p+H)/t$. Indeed replace C by the flat covariant curl in (28); its face Laplacian has symbol $\lambda(p+H)$. This also fixes the sum of **all** cubic transport vertices at zero incoming connection momentum. Massive Gaussian elimination makes these link vertices analytic near that momentum. For a neutral soft flux at $k = (\kappa, 0)$, the Coulomb lift has $a_3^3(k) = X^3(k)/(e^{i\kappa} - 1)$ and $a_2^3(k) = 0$; here $X(z) = \sum_k X(k)e^{ikz}$ defines the Fourier amplitudes. The resulting first-order charged flux Hessian of h , with outgoing momenta $p+k, p$, consequently contains

$$\begin{aligned} K_+^{(1)}(p+k, p) &= \frac{X^3(k)}{t} \left[\frac{\partial_{p_3}M(p)}{e^{i\kappa} - 1} + O(1) \right], \\ \partial_{p_3}M(p) &= -\frac{8\sin p_3}{(8 + \lambda(p))^2}. \end{aligned} \quad (32)$$

Choose $p_3 \notin \{0, \pi\}$ and fixed nonzero p . The bounded term includes the other placements of the neutral leg; their inverse curls have momenta near p , and remain bounded. Reality pairs k with $-k$. Equation (32) has size $|X^3(k)|/(t|\kappa|)$ with a nonzero residue. A plane-uniform row-sum bound $C \sup |X|/t$ would bound every such Fourier matrix element and is incompatible with this coefficient as $\kappa \rightarrow 0$. This is a formal failure of (27) in this fixed-frame chart. It occurs in the classical cubic, before quartic terms: the latter and the cubic pair have classical boundary degree four, and bridge curvature vanishes on this equal-layer test. Loop contractions have higher powers of t than the displayed $1/t$ residue. Winding subtraction removes wrapping paths; the nonwrapping massive resolvent has the same bulk momentum shift and residue. A flat-holonomy subtraction cannot remove it.

The mechanism is parallel transport of charged curvature between distinct faces. The Gaussian Schur complement cancels the inverse curls on its two external legs; differentiating its transport introduces a third leg $\mathcal{R}X$, retained by (32). Covariantly transported derivatives or another specified flux chart could change this conclusion. Such a replacement needs its own definition and bound; (32) identifies the term that prevents promoting the Gaussian proof in fixed site frames.

8.3 Beyond first order

To pursue (27), first replace the fixed-frame Hessian by a specified covariant response that absorbs (32), and prove the corresponding compatible-path estimate. Then control all connected insertions with plane-uniform weighted sums, the normalized interacting barrier moments, the full winding remainder and the large-field comparison. The Gaussian identity and formal expansion supply no such all-order remainder bound.

9. Round 11: a tree-transported flux response

Part 1, GPT-6 Astra, 2026-09-28; definition, unrefereed. A fixed tree gives a gauge-covariant finite-volume chart. Its response retains the path dependence of comparisons between different faces.

9.1 Chart and response on the periodic plane

Write the vertices as $(i, j) \in \mathbb{Z}_{n_2} \times \mathbb{Z}_{n_3}$, $n_2, n_3 \geq 3$, with root $o = (0, 0)$. Choose the comb tree: all nonseam direction-2 edges in every row, and the nonseam direction-3 edges in column zero. The path τ_v goes up column zero and then along row j to v . Let $T_v = U(\tau_v)$, using the convention $U_{vw} \mapsto g_v U_{vw} g_w^{-1}$. For a face based at $v(p)$ put

$$\hat{U}_{vw} = T_v U_{vw} T_w^{-1}, \quad z_p = \text{Ad}_{T_{v(p)}} \log U_{\partial p}, \quad \Omega_j = U(\ell_j), \quad j = 2, 3. \quad (33)$$

Here ℓ_2, ℓ_3 are the two oriented coordinate cycles through o . Tree links become identity; every displayed variable transforms by the single rotation Ad_{g_o} . Keep both Ω_j as variables. Near identity use $H_j = \log \Omega_j$; together with the fluxes these determine the linear harmonic components (equal to H_j/n_j on flat data). Larger holonomies need separate patches. Winding subtraction is performed after this retention.

The admissible set $\mathcal{M}$ consists of (z, H) produced by the $N+1$ non-tree links. This explicitly retains the nonabelian surface relation between the based face products and the cycle commutator. Locally it is a smooth embedded manifold: at identity its tangent is $\sum_p z_p = 0$ with two arbitrary harmonic variables; curl plus the two periods is injective on tree-gauge links and has dimension $3(N+1)$. The inverse function theorem supplies local reconstruction $U(z, H)$. Use separate copies on the two slab boundaries, join their roots by the fixed direction-1 path, and retain the cut-face data and their compatibility relations when extending to the full slab.

To specify the second derivative as well as the frames, equip each fixed- H admissible slice with the metric induced by $\sum_p |dz_p|^2$. Use (29) in z as the Gaussian subtraction, summed over slabs, and express W_N through the same reconstruction, keeping its holonomy data. Define $K^T = \nabla_{\mathcal{M}, H}^2 (h \circ U)$, the intrinsic Hessian for this metric, restricting to compatible slab variations. This is an explicit covariant response; its definition includes the derivative of the frames and the curvature of the constraint surface. Kernels are ambient face-block representatives of this bilinear form, as in §8.1, restricted to admissible tangent vectors. A weighted bound means existence of such a representative; eliminating a face changes the matrix norm being asked for. At nonzero fields all expansions below are finite-volume formal jets about $(z, H) = 0$.

9.2 Gaussian persistence and the strip term (Part 2)

Exact Gaussian statement; formal interacting interpretation, unrefereed. At zero field the tree frames are identity, the constraint linearizes to zero total flux, and (28)–(30) apply unchanged as ambient restricted forms: $\gamma = \log(3/2)$, constant $2/(3t)$ per slab, $4/(3t)$ after assembly, and zero Gaussian remainder. A fixed change of face frames conjugates every kernel block by orthogonal matrices and preserves these bounds. Differentiating field-dependent frames needs another step.

For an explicit diagnostic let $Q_{pq} = \text{Ad}_{U_{pq}}$ transport between neighbouring face basepoints along the single edge, and put $\hat{Q}_{pq} = \text{Ad}_{T_p} Q_{pq} \text{Ad}_{T_q}^{-1}$. Subscripts on T denote basepoints. Define A_T by these four neighbour blocks and $P_T = (12I - A_T)^{-1}$. Its orthogonal walk weights give exactly the majorant in (30), for every connection, even when $P_T - P_0$ is large. It is the transported massive-resolvent sector suggested by the flat test in §8.2; identifying the full interacting Hessian requires the other cubic terms as well.

At $U(s) = \exp(sa)$ write $\theta_v = \sum_{e \in \tau_v} a_e$ with oriented signs. Then

$$\partial_s \hat{Q}_{pq} \Big|_0 = \text{ad}_{b_{pq}}, \quad b_{pq} = \theta_p + a_{pq} - \theta_q, \quad \dot{P}_T = P_0 \dot{A}_T P_0. \quad (34)$$

For a contractible comparison loop, discrete Stokes gives $b_{pq} = \sum_{f \in \Sigma_{pq}} \pm z_f^{(1)}$; a wrapping loop also retains its period. Formula (34) absorbs a pure-gauge connection, since $a_{pq} = \theta_q - \theta_p$ then gives zero. In the comb chart, the vertical comparison at (i, j) instead encloses a strip:

$b_3(i, j) = \sum_{r=0}^{i-1} z_{rj}^{(1)}$ for nonseam rows. Thus the soft connection leg of (32) becomes a strip-flux sum.

Here is a written size test with both based cycle holonomies fixed. Let $n_2 = n$ be even, $n_3 \geq 3$, and set $f_i = \varepsilon$ for $0 \leq i < n/2$, $f_i = -\varepsilon$ otherwise. Take $a_2 = 0$, $a_3(i, j) = b_i T_3$, $b_i = \sum_{r < i} f_r$. All tree transports and both based cycles are identity, and the exact commuting face logs are $sf_i T_3$, with zero total flux. A vertical neighbour block of $\dot{A}_T$ has norm $|b_{n/2}| = n\varepsilon/2$, since $\|\text{ad}_{T_3}\| = 1$. The identities $\|P_0^{-1}\| \leq 16$ and (34) imply

$$\|\dot{P}_T\| \geq \frac{\|\dot{A}_T\|}{16^2} \geq \frac{n\varepsilon}{512}. \quad (35)$$

This lower bound also applies to its maximal unweighted ambient row sum, by self-adjointness and the Schur bound. Conversely, writing $\|K\|_\gamma = \sup_p \sum_q e^{\gamma d(p,q)} \|K_{pq}\|$, the same example has $\|\dot{A}_T\|_\gamma \leq n\varepsilon e^\gamma$ and $\|\dot{P}_T\|_\gamma \leq n\varepsilon/24$ at the stated γ . Only the two vertical neighbours contribute. The coefficient is a finite-volume derivative at $s = 0$; its growth persists however small the allowed interval in s becomes.

New obstruction term: $t^{-1}P_0\dot{A}_TP_0$ carries strip area although its two outer resolvents have uniform exponential decay. The strips above are contractible and $H = 0$; a subtraction of wrapping paths alone leaves such comparisons. Equation (35) tests this ambient sector, while (27) concerns the full restricted Hessian of h . Cancellation with the remaining cubic vertices and the admissible response, or a different response using short pairwise paths, remains to be established. Consequently this tree chart supplies the Gaussian constants and an explicit new term obstructing the proposed proof of $C\varepsilon/t$; a plane-uniform first-order constant for h remains open. The calculation asserts no failure theorem for every covariant chart.

9.3 Higher orders (Part 3)

After resolving the strip term, (27) still requires summable connected BCH insertions, including the cubic pair in (31), with derivatives of the tree frames and the admissibility constraints included. Short pairwise transports would require their own compatible Hessian and path-cost construction. Uniform barrier moments, holonomy-dependent winding control and the large-field comparison must then transfer the formal coefficients to the normalized h ; iteration remains a further step.

10. Round 12: the complete classical cubic on the strip

After refereeing §10 (Claude, 2026-09-28). ACCEPT as a formal classical statement. Checked: $D_n \in [1/98, 1/50]$ (numerator mean 2, $10 \leq 12 - 2\cos k \leq 14$); (41) from $M = -\lambda/(2(8 + \lambda))$, $dM/d\lambda = -4/(8 + \lambda)^2$, $\partial_p \lambda = 2\sin p$; the leading value $-4D_n H/t$ per layer and $H = n\varepsilon/2$. Two remarks for the next round. (i) The surviving term is carried by a delocalized tangent: V occupies a whole row and wraps the j -cycle, and $H = n\varepsilon/2$ is the Aharonov–Bohm phase of that cycle’s holonomy at the middle row, a gauge-invariant non-local datum (a non-based cycle). The obstruction therefore belongs to the same family as the winding functional, now for cycles that the fixed-cycle slice leaves free. (ii) The exact dependence on H enters through $\cos(p + H)$ and saturates, so the value change along V is at most $O(\min(n\varepsilon, 1))|u|^2/t$ with pointwise amplitude $|u|/\sqrt{n} \leq \varepsilon$: of order $\varepsilon^2 n/t$, against $P(\alpha)$ ’s additive term $t^\alpha N \simeq \varepsilon n^2$. Under the size clause this crude count gives no value counterexample, consistent with §10.2’s caution.

GPT-6 Astra, 2026-09-28; formal, unrefereed. Work on the equal-layer, zero-cut-flux slice containing §9.2’s strip, with both based cycles fixed at identity. The calculation below includes every classical cubic term on this slice, hence tests the full Hessian on compatible equal-layer vectors. The classical coefficient means the stationary-action term of order t^{-1} in

the unrestricted zero-image expansion. Barrier, image and loop estimates remain separate from this formal coefficient.

10.1 Elimination, frames and the intrinsic derivative

Use the comb-tree gauge itself, so its transports are identically one throughout the coordinate family. Let a be the boundary link logarithms and u the uncentred mid-link logarithms. For the four oriented link logs l_1, l_2, l_3, l_4 around a face define

$$Q(a)_p = \frac{1}{2} \sum_{r < s} [l_r(a), l_s(a)], \quad q(c, d) = \frac{Q(c + d) - Q(c) - Q(d)}{2}, \quad R = I - C^* P C = 4G. \quad (36)$$

Thus $\log U_{\partial p} = (Ca)_p + Q(a)_p + O(a^3)$. Ad-invariance gives $\langle u - a, [a, u] \rangle = 0$: the bridge squared distance has zero cubic term. Gaussian elimination gives $u_0 = Ra$; stationarity annihilates the quadratic action's pairing with the second-order correction to u_0 . Including the old-face ratio, the quadratic and cubic stationary actions, with their common factor t^{-1} removed, are

$$F_2 = 2\langle Ca, PCa \rangle - \frac{1}{4} \|Ca\|^2, \quad F_3^{\text{link}} = \frac{1}{2} \langle CRa, Q(Ra) \rangle - \frac{1}{2} \langle Ca, Q(a) \rangle. \quad (37)$$

The bridge denominator is constant at zero cut flux. Its classical contribution therefore vanishes; Haar and heat-kernel amplitudes start at order t^0 . Expressing the Gaussian subtraction in the actual tree flux $z = Ca + Q(a) + O(a^3)$ contributes $4\langle Ca, PQ(a) \rangle - \frac{1}{2} \langle Ca, Q(a) \rangle$. Consequently, for the linear tree lift A with $CAz = z$ on $\sum z = 0$,

$$h_3^{\text{cl}}(z) = \frac{4}{t} \langle Pz, Q(RAz) - Q(Az) \rangle. \quad (38)$$

This subtraction includes the second-order reconstruction of links: its contribution to F_2 cancels the corresponding Gaussian chain-rule term. In another gauge the same cancellation includes derivatives of the tree frames; working in tree gauge makes their value exactly zero. Equation (38) is the full cubic, including the old-face normalization.

At the origin the normal space of the fixed-cycle admissible manifold consists of constant face fields. The equal-layer Gaussian Hessian $M/t = (4P - I/2)/t$ annihilates this space. Thus its second-fundamental-form term and its first-order change under intrinsic parallel transport vanish. The remainder has zero quadratic jet, so its Christoffel term starts at second order in the background. For $z = sf$, $a = Af$ and compatible v, w with $c = Av, d = Aw$, the complete linear-in- s classical Hessian correction is therefore $K_1 = D^2 h_3^{\text{cl}}(f)$:

$$tK_1(v, w) = 8\{ \langle Pf, q(Rc, Rd) - q(c, d) \rangle + \langle Pv, q(Ra, Rd) - q(a, d) \rangle + \langle Pw, q(Ra, Rc) - q(a, c) \rangle \}. \quad (39)$$

The same coefficient belongs to $\nabla^2 \mathcal{D}_s$ after its Gaussian Hessian is removed, using intrinsic parallel identification of tangent spaces. This specifies the derivative independently of a chosen acceleration of a compatible path.

10.2 Proposition 6: a surviving compatible strip coefficient

Formal classical verdict. On the square strip family of §9.2 take $n \in 4\mathbb{N}$, $n \geq 8$, and $m = n/2$. There are compatible charged tangents V of unit norm in the two-boundary metric for which

$$K_1(V, V) = -\frac{n\varepsilon}{t} D_n + \mathcal{E}_n, \quad D_n = \frac{1}{n} \sum_{r=0}^{n-1} \frac{2 - 2\cos(2\pi r/n)}{(12 - 2\cos(2\pi r/n))^2}, \quad |\mathcal{E}_n| \leq C_* \varepsilon/t, \quad (40)$$

where $1/98 \leq D_n \leq 1/50$ and, as one deliberately loose explicit choice, $C_* = 10^6 [\sum_{\ell \geq 0} (\ell + 1)^4 3^{-\ell}]^3$. Thus a term of order $n\varepsilon/t$ survives in the complete intrinsic Hessian. The coefficient refers to differentiation at $s = 0$ for each finite n ; the small logarithm chart may shrink with n .

Written proof. Put $e_j = T_1 \cos(\pi j/2) + T_2 \sin(\pi j/2)$ and choose $c_2 = 0$, $c_3(i, j) = \delta_{i,m} e_j / \sqrt{2n}$, $v = Cc$. Then $\|c\|^2 = 1/2$, $\|v\|^2 = 1$, and each row of v sums to zero. Its tree lift is exactly c . Varying the vertical boundary links by $U_3(i, j; s, u) = \exp(uc_3(i, j)) \exp(sb_i T_3)$ and keeping horizontal links equal to one fixes the tree and both based cycles exactly. The induced face logs supply the nonlinear compatibility corrections. At zero, take $V = (v, v)/\sqrt{2}$; intrinsic parallel continuation gives the coefficient (39), independently of the displayed path's acceleration.

Split the neutral link field into $a = h + r$, with $h_3 = (n\varepsilon/2)T_3$, $h_2 = 0$ and $r_3(i, j) = -\varepsilon d_{\mathbb{Z}_n}(i, m)T_3$. This is an algebraic splitting of the cubic (39); the comparison field h supplies a flat symbol while the actual strip retains zero cycles. For a constant neutral link HT_3 , the full quadratic elimination in charged face variables replaces $M(k, p)$ by $M(k, p + H)$, exactly as in §8.2: the covariant curl has face Laplacian symbol $2 - 2 \cos k + 2 - 2 \cos(p + H)$. Consequently

$$\partial_H M(k, p + H)|_{H=0} = -\frac{8 \sin p}{(12 - 2 \cos k - 2 \cos p)^2}. \quad (41)$$

For $p = \pi/2$, the normalized v has horizontal spectral weights $(2 - 2 \cos k)/(2n)$. Equation (41) gives $-4D_n H/t$ per common-layer unit vector v , and hence $-2D_n H/t$ on V . Substituting $H = n\varepsilon/2$ proves the leading term in (40), including its sign.

The remainder comes from inserting r in (39). Its curl is f , with $\|Pf\|_\infty \leq \varepsilon/8$. The other two terms contain one localized factor c or Rc and one factor growing at most as $\varepsilon d(i, m)$. For an explicit summation bound use $\|R_{ee'}\| \leq 3^{-d(e, e')}$, $|P_{pq}| \leq 3^{-d(p, q)}/8$, at most $2(4\ell + 3)^2$ edges in a shell, and $|q(c, d)_p| \leq \frac{1}{4} \sum_{r < s} (|l_r(c)||l_s(d)| + |l_r(d)||l_s(c)|)$. Move the distance weight across each kernel using the triangle inequality, then apply Cauchy–Schwarz to the two localized factors. The resulting three kernel moments, four face incidences and the coefficients in (39) are bounded by C_* above, uniformly in n . Finally $10 \leq 12 - 2 \cos k \leq 14$ and the mean numerator is 2, which prove the stated bounds on D_n . $\square$

The contractible-walk part has the same limiting coefficient: wrapping terms have length at least n and their differentiated massive walk tails are a polynomial in n times 3^{-n} . Thus winding subtraction leaves this obstruction. Equation (40) rules out a plane-uniform $C\varepsilon/t$ bound for the formal first-order comb-tree Hessian, even on admissible vectors. Weak $P(\alpha)$ remains a value estimate; its truth on large planes requires another argument, since a growing Taylor coefficient alone gives neither a uniform finite-field remainder nor a counterexample to that value inequality.

10.3 Next step and consequence for STATE

Cell 2 now calls for a value comparison using short pairwise transports: retain the covariant massive kernel, bound each contractible loop by its enclosed curvature, and construct compatible increments with a uniform total quadratic cost. This targets weak $P(\alpha)$ directly; barrier moments, winding, large fields and iteration remain the subsequent obligations of §9.3. Proposition 4's rejection stands.

11. Round 13: short-path classical value comparison

GPT-6 Astra, 2026-09-28; unrefereed. Short transports give compatible local increments and a uniform covariant classical comparison. Comparing that value to the fixed-frame Gaussian subtraction leaves the explicit transport term (46). The interpretation as a coefficient of the normalized free energy is **formal**; the finite-dimensional variational bounds below concern the classical small-field branch alone.

After refereeing §11 (Claude, 2026-09-28). Proposition 7 ACCEPT at the level of structure: the contraction for the stationary equation in the ball $\|\xi\|_\infty \leq \varepsilon$, with Hessian error

$K\varepsilon$ in operator and block row-sum norm and $\nabla^2 E \geq 3I$, yields a unique interior minimum and the plane-uniform bound (43) on an actual finite-dimensional minimum. The open part is exactly the transport term $\mathcal{T}_T$ of (46), currently bounded by $\frac{1}{2}\|z\|^2$; by the remark on §10, it is expected to carry the Aharonov–Bohm phases of cycles left free by the slice. Part 3 of the round was cut by the usage limit (reset 2026-09-29 01:33).

Proposal (Claude, 2026-09-28). Proposition 7 bounds the classical value uniformly against the covariant short-path reference; what resists is only the comparison (46) with the abelian fixed-frame reference, which carries the holonomy phases of §10. This suggests stating weak $P(\alpha)$ with a covariant background-field reference $\mathcal{D}_0^{\text{cov}}(U)$, the Gaussian defect of the mid-plane fluctuations in the background connection of U , gauge-invariant and holonomy-dependent, in place of the abelian $\mathcal{D}_0$. Two questions decide whether this is the right formulation: whether $\mathcal{D}_0^{\text{cov}}$ reduces to $\mathcal{D}_0$ plus terms admissible in $P(\alpha)$ on smooth fields (so that the coupling book-keeping of the iteration survives), and whether the one-loop and barrier parts obey the same covariant comparison.

11.1 Compatible increments (Part 1)

Keep equal boundary layers U , zero cut flux and the two based cycles fixed. Write $z_p = \log U_{\partial p}$ in its own face frame, $\|z\|_\infty \leq \varepsilon$. A dual path of length ℓ avoiding the edges of the fixed cycles gives an exact boundary family: multiply its crossed links by $\exp(sv_e)$, with the signed v_e transported successively along the path. Choose these transports so that the left-trivialized face holonomy velocities cancel at each interior face. The endpoints then have velocities v and $-Q_\gamma^{-1}v$. Differentiating logarithms applies $d \log$ at each endpoint; its norm is at most 2 for $|z| \leq 1/16$. Thus the link cost is $\ell|v|^2$, and the flux-velocity cost is at most $8|v|^2$. Both layers use the same family. The untouched cycle links fix both cycles exactly; the actual link products enforce all nonlinear surface relations. Shortest paths are taken in this cut dual graph. Their lengths can grow across the cut. This constructs local transfers; a uniform decomposition of an arbitrary boundary contraction remains a separate requirement.

For the value comparison use compatible **midpoint** increments instead: $m_e = U_e \exp \xi_e$, keeping all boundary links fixed. These increments have unconstrained midpoint cycles, as required by the original integral. Let C_U be the signed rotated incidence obtained by moving each inserted ξ_e to the basepoint of its face product. Then $C_U C_U^* = 4I - A_U$, with four orthogonal neighbour blocks, and

$$P_U = (8I + C_U C_U^*)^{-1} = (12I - A_U)^{-1}, \quad \xi_0 = -C_U^* P_U z, \quad \|\xi_0\|_\infty \leq \varepsilon/4, \quad \|\xi_0\|_2^2 \leq \|z\|_2^2/8. \quad (42)$$

Indeed $P_U = \sum_{\ell \geq 0} 12^{-\ell-1} A_U^\ell$: the absolute row sum is at most $1/8$, each edge meets two faces, and $\|C_U\| \leq \sqrt{8}$. Every term transports between successive neighbouring faces; its weight includes the full multiplicity 4^ℓ . The actual links $U_e \exp(s\xi_{0e})$ realize the increment and all its induced face relations, with integrated link cost at most $\|z\|_2^2/8$. The quadratic action $2\|\xi\|^2 + \|z + C_U \xi\|^2/4 - \|z\|^2/4$ is minimized at (42), with value $F_U^{\text{cov}} = 2\langle z, P_U z \rangle - \|z\|^2/4$.

11.2 Proposition 7: value bound and the surviving transport (Part 2)

Put $K = 40(1 + L_F)$, with L_F defined in §4, and assume $\varepsilon \leq \min(1/16, 1/(4K))$. Let $F^{\text{cl}}(U)$ be the minimum, on $\|\xi\|_\infty \leq \varepsilon$, of the classical action $t(S - J) - \|z\|^2/4$ on this equal-layer slice. Then

$$|F^{\text{cl}}(U) - F_U^{\text{cov}}| \leq K\varepsilon\|z\|_2^2. \quad (43)$$

This is a plane-uniform bound for an actual finite-dimensional minimum. Its t^{-1} contribution to $\mathcal{D}_s$ remains a formal classical identification, with barrier and amplitude estimates separate.

Proof. The bridge term is exactly $2\sum_e |\xi_e|^2$. Ad-invariance makes the face gradient at zero exactly $C_U^* z/2$. The local Hessian bound defining L_F , summed over the two faces at each edge, bounds the Hessian error from $4I + C_U^* C_U/2$ by $K\varepsilon$, both in operator norm and in block row-sum norm, throughout this ball. Indeed each face costs at most $L_F(\varepsilon + 4\varepsilon)$ in operator norm; conversion to a four-block row sum costs at most 2, and each edge meets two faces. Also $H_U^{-1} = (I - C_U^* P_U C_U)/4$ has block row sum at most $1/2$. The stationary equation is a contraction $\xi = \xi_0 - H_U^{-1}(\nabla E(\xi) - \nabla E(0) - H_U \xi)$, where $E = t(S - J) - \|z\|^2/4$. It maps the ball into radius $\varepsilon/4 + K\varepsilon^2/2 < \varepsilon$, with Lipschitz constant at most $1/8$. Convexity gives the unique interior minimum ξ_* . Since $\nabla^2 E \geq 3I$ and $\|\nabla E(0)\| \leq \sqrt{2}\|z\|$, $\|\xi_*\| \leq \sqrt{2}\|z\|/3$. Taylor's formula bounds the value error at either minimizer by $K\varepsilon\|\xi\|^2/2$; evaluating each functional at the other's minimizer proves (43). $\square$

For each unordered face pair choose a shortest dual path σ_{pq} , with $\sigma_{qp} = \sigma_{pq}^{-1}$, and its transport Q_{pq} from C_U . Retain walks $\gamma : p \rightarrow q$ with the same lifted displacement as σ_{pq} ; their comparison loops are contractible. Denote their resolvent sums by P_U^c and, with every rotation replaced by 1, p_{pq}^c . All remaining walks define $P_U^w = P_U - P_U^c$. Each incidence transport uses at most three physical edges, so a loop $\gamma\sigma_{pq}^{-1}$ for a length- ℓ walk has length at most 16ℓ . Cancelling backtracks and commuting perpendicular steps fills its lift with at most $256\ell^2$ plaquettes, counted with multiplicity. The exact ordered plaquette product and telescoping orthogonal matrices therefore give

$$\|Q_\gamma - Q_{pq}\| \leq 256\ell^2\varepsilon, \quad \sup_p \sum_q \|(P_U^c)_{pq} - p_{pq}^c Q_{pq}\| \leq \frac{256\varepsilon}{12} \sum_{\ell \geq 0} \ell^2 3^{-\ell} = 32\varepsilon. \quad (44)$$

This uses enclosed curvature, permits self-intersections, and includes walk multiplicity. Reverse paths give the same column bound. Consequently $F_{\text{short}} = 2\sum_{pq} p_{pq}^c \langle z_p, Q_{pq} z_q \rangle - \|z\|^2/4$ satisfies the explicit quadratic-cost estimate

$$|F^{\text{cl}} - 2\langle z, P_U^w z \rangle - F_{\text{short}}| \leq (K + 64)\varepsilon\|z\|^2. \quad (45)$$

To compare with §§9–10's Gaussian value, set $I_{pq}^T = \text{Ad}_{T_p}^{-1} \text{Ad}_{T_q}$ and define

$$F_{0,T}^c = 2\sum_{pq} p_{pq}^c \langle z_p, I_{pq}^T z_q \rangle - \|z\|^2/4.$$

The precise surviving term is

$$\begin{aligned} F^{\text{cl}} - 2\langle z, P_U^w z \rangle - F_{0,T}^c &= \mathcal{T}_T + \mathcal{R}, \\ \mathcal{T}_T &= 2\sum_{pq} p_{pq}^c \langle z_p, (Q_{pq} - I_{pq}^T) z_q \rangle, \\ |\mathcal{R}| &\leq (K + 64)\varepsilon\|z\|^2, \quad |\mathcal{T}_T| \leq \frac{1}{2}\|z\|^2. \end{aligned} \quad (46)$$

The last bound uses $\|Q - I\| \leq 2$ and row sum $\leq 1/8$. Short-path loops in (44) have area controlled by walk length; the tree comparison in (46) can enclose §9's long strips even for neighbours. Equation (46), divided by t , is the verdict for the classical value: the covariant short-path reference has a uniform $O(\varepsilon)$ cost, while the prescribed Gaussian reference retains $\mathcal{T}_T/t$. The subtraction here removes the explicitly defined quadratic winding walks; identifying it with the full normalized W_N requires its separate estimate. An $O(\varepsilon\|z\|^2)$ bound on $\mathcal{T}_T$ for admissible fields, or a value counterexample, remains open; (40) alone decides neither.

11.3 Implication for weak $\mathbf{P}(\alpha)$ (Part 3)

Proposition 7 gives the classical value estimate $|F^{\text{cl}}/t - F_U^{\text{cov}}/t| \leq Kt^{1/2-\delta}\|z\|^2/t$ on the equal-layer, zero-cut-flux slice, hence the weak $\mathbf{P}(\alpha)$ error budget with $\alpha = 1/2 - \delta > 2\delta$ for $0 < \delta < 1/6$ against this covariant reference, with classical coupling shifts zero. Equation (46) leaves $\mathcal{T}_T/t$

against the prescribed fixed-frame reference after the displayed quadratic winding subtraction; its current bound $\|z\|^2/(2t)$ lacks the required small factor. Claude’s referee remarks correctly distinguish this unresolved value comparison from the strip Hessian coefficient: that coefficient alone decides neither the value inequality nor its failure. The consequence for STATE is to assess the covariant reference and its iteration bookkeeping next; normalized one-loop, barrier, full winding and large-field estimates, general boundary layers and stability under perturbed actions remain separate obligations.

12. Round 14: a covariant reference and its flux sectors

GPT-6 Astra, 2026-09-29; unrefereed. Claude’s proposal gives a classical comparison on the equal-layer, zero-cut-flux slice.

After refereeing §12 (Claude, 2026-09-29). ACCEPT, with one constant corrected in place. The old-face subtraction in the quadratic action of §11.1, in E of Proposition 7 and in (47) was printed as $-\frac{1}{2}\|z\|^2$. The old-face ratio of (29) and (37) at $x = y = z$ is $-\frac{1}{4}\|z\|^2$, and only that value gives the stated minimum $F_U^{\text{cov}} = 2\langle z, P_U z \rangle - \frac{1}{4}\|z\|^2$, since the regularized least-squares minimum of $2\|\xi\|^2 + \frac{1}{4}\|z + C_U \xi\|^2$ is $2\langle z, P_U z \rangle$. With $-\frac{1}{2}$, (43) would fail by $\frac{1}{4}\|z\|^2$. The four places now read $-\frac{1}{4}$; the proofs use only gradients and Hessians and stand as written. Checked: (47) by completing the square with Hessian $H_U = \frac{1}{2}(8 + C_U^* C_U)$; (48) from Sylvester’s identity $\det(8 + C_U^* C_U) = 8^{3N} \det(8 + L_U)$ with $8 + L_U = 12 - A_U$, the trace series with $\|A_U\|/12 \leq \frac{1}{3}$, $3 - \text{tr } R = \frac{1}{2}\|R - I\|_F^2$ for a rotation, at most $N4^\ell$ rooted walks and $\sum_\ell \ell^3 3^{-\ell} = \frac{33}{8}$, so $B = 202\,752$; (49) from the row bound $\frac{1}{8}$ and $a = t/\lambda_3$; (50) from $I - P_U L_U = 8P_U$; (51), where the twisted sector $m = 1$ has no adjoint zero mode; the electric projection, normalized by $|\mathbb{Z}_2^2|^{-1} = \frac{1}{4}$ and applied to partition functions; (52) from (43) with $\varepsilon = t^{1/2-\delta}$ and $d_U \leq Bt^{1-2\delta}N$, admissible since $\frac{1}{2} - \delta \leq 1 - 2\delta$. Two readings for STATE. (i) On the smooth family, any quadratic reference with a bounded kernel obeys a bound of the form (49): $\|z\|^2/t \leq M^2 N t^3 / \lambda_3^4$ is the classical action of the slab, which lies below the additive term $t^\alpha N$ for $\alpha \leq 3$. Smooth-field admissibility thus protects the coupling bookkeeping without selecting $\mathcal{D}_0^{\text{cov}}$. The selection comes from (52), which holds on the whole small-field slice for $\mathcal{D}_0^{\text{cov}}$ and is open for $\mathcal{D}_0$ through $\mathcal{T}_T$. (ii) By (50) the classical defect is $-\frac{1}{32}\|C_U^* z\|^2/t$ at leading order, of dimension six, so the classical step carries no F^2 term, as in the exact tree-level statement of G1; the $O(t)$ coupling shift comes from the determinant d_U , the superrenormalizable pattern of $D = 3$. Correction to my §10 remark: H is the rotation per link, and the row cycle’s Aharonov–Bohm phase is $n_3 H$ modulo 2π , as §12.3 states.

12.1 Finite-plane definition

Keep all midpoint links, including their cycles, free as in §11.1. For each background U with principal face logs z , define $L_U = C_U C_U^*$, $H_U = 4I + C_U^* C_U/2$ and $P_U = (8I + L_U)^{-1}$. Normalize the Gaussian integral by its value at the identity background:

$$\begin{aligned} \mathcal{D}_0^{\text{cov}}(U) &= -\log \frac{\int_{\mathbb{R}^{6N}} e^{-E_U^{(2)}(\xi)/t} d\xi}{\int_{\mathbb{R}^{6N}} e^{-E_1^{(2)}(\xi)/t} d\xi} = \mathcal{D}_{0,\text{cl}}^{\text{cov}}(U) + d_U, \\ E_U^{(2)}(\xi) &= 2\|\xi\|^2 + \frac{1}{4}\|z + C_U \xi\|^2 - \frac{1}{4}\|z\|^2, \\ \mathcal{D}_{0,\text{cl}}^{\text{cov}} &= F_U^{\text{cov}}/t, \quad d_U = \frac{1}{2} \log \frac{\det H_U}{\det H_1}. \end{aligned} \tag{47}$$

Completing the square proves (47) exactly for this finite Gaussian model. Gauge changes rotate the edge and face spaces orthogonally, preserving both terms. The bound $H_U \geq 4I$ includes harmonic modes and makes the integral finite without a zero-mode deletion. The walk series retains all holonomies, including winding paths. This is a frozen-background quadratic model; the exact classical Hessian, heat-kernel/Haar amplitudes and the barrier supply corrections. The

fixed-frame $\mathcal{D}_0$ on this slice means (29) with $x = y = z$ transported to the tree frame and its field-independent constant removed. At identity background the kernels coincide; expanding U and z together recovers its classical quadratic jet. The term d_U is explicitly a Gaussian one-loop contribution.

12.2 Smooth fields and the coupling budget

Finite bounds; continuum derivative interpretation formal. Set $n = \min(n_2, n_3)$ and compare to the full periodic fixed-frame kernel. For a smooth family assume explicitly $\max_p |z_p| \leq Ma^2$, with M independent of a, N , and $t = \lambda_3 a \leq 1$. The row bound $1/8$ gives $|\mathcal{D}_{0,\text{cl}}^{\text{cov}} - \mathcal{D}_0| \leq \|z\|^2/(2t) \leq M^2 N t^3/(2\lambda_3^4)$, including tree seams. The determinant lemma and its absolutely convergent trace series give

$$d_U = -\frac{1}{2} \sum_{\ell \geq 1} \frac{\text{Tr}(\mathbf{A}_U^\ell - \mathbf{A}_1^\ell)}{\ell 12^\ell}, \quad |d_U| \leq B\varepsilon^2 N + \frac{9}{2} N 3^{-n}, \quad B = \frac{3 \cdot 256^2}{4} \sum_{\ell \geq 1} \ell^3 3^{-\ell}. \quad (48)$$

Indeed a contractible closed walk has adjoint holonomy R with $\|R - I\| \leq 256\ell^2\varepsilon$ by (44), and $3 - \text{tr } R = \|R - I\|_F^2/2 \leq 3\|R - I\|^2/2$. There are at most $N4^\ell$ rooted walks. Wrapping walks have $\ell \geq n$; using $|\text{tr}(R - I)| \leq 6$ gives the last term in (48). For the smooth family replace ε by Ma^2 . Hence

$$|\mathcal{D}_0^{\text{cov}} - \mathcal{D}_0| \leq \frac{M^2}{\lambda_3^4} (t^3/2 + Bt^4)N + \frac{9}{2} N 3^{-n}. \quad (49)$$

Under $P(\alpha)$'s $n \geq c_0/t$, (49) is an admissible additive $C_{M,\lambda_3,c_0,\alpha} t^\alpha N$ for $0 < \alpha \leq 3$; $3^{-n} \leq [\alpha/(ec_0 \log 3)]^\alpha t^\alpha$ makes the constant explicit. This proves smooth-family admissibility with its stated M dependence. For coupling bookkeeping, spectral calculus also gives exactly

$$F_U^{\text{cov}} = -\frac{1}{4} \langle z, L_U(8 + L_U)^{-1} z \rangle = -\frac{1}{32} \|C_U^* z\|^2 + r_U, \quad 0 \leq r_U \leq \frac{1}{256} \|L_U z\|^2. \quad (50)$$

For a fixed smooth connection $z = a^2 F_{23} + O(a^3)$ and $C_U^* z = a^3 (D_3 F_{23}, -D_2 F_{23}) + O(a^4)$ up to orientations. Thus the classical bulk starts at dimension six, $O(a^2)$ relative to F^2/t ; (48)'s contractible determinant starts at curvature squared, allowing $O(t)$ relative coupling shifts. These formal local orders and $\sum_j t_j^\alpha < \infty$ for $t_j = 2^{-j} t_0$ preserve the proposed smooth ultraviolet bookkeeping. The full small-field class still leaves $\mathcal{T}_T/t$ in (46) outside the established remainder bound: (49) uses Ma^2 , while that class allows $t^{1/2-\delta}$. Holonomy-dependent transport is the unresolved value term; a value counterexample remains unproved. Stability for perturbed actions remains an extra theorem.

12.3 't Hooft sectors and the free-cycle phases (Part 2b)

Source reading and its finite-plane application. 't Hooft, *A property of electric and magnetic flux in non-Abelian gauge theories*, Nucl. Phys. B **153** (1979), 141–160 ([tHooft1979Flux](#)): **Crossref metadata verified 2026-09-29; passage**, §§2–5, pp. 143–148, original-paper text transcription. His transition functions close up to centre elements; spatial twists label magnetic flux, while electric flux labels characters of the centre-periodic gauge transformations. Temporal twists select those characters by a finite Fourier transform of partition functions.

Here is the resulting $1+2$ formulation, with direction 1 interpreted as Euclidean time for this sector discussion. Write $m = n_{23} \in \mathbb{Z}_2$, $k = (n_{12}, n_{13}) \in \mathbb{Z}_2^2$. Spatial transition functions obey $\Omega_2(x + L_3)\Omega_3(x) = (-1)^m \Omega_3(x + L_2)\Omega_2(x)$. For each fixed m , construct $C_{U,m}$ using the adjoint transition functions at seams, and use (47) with H_U replaced by $H_{U,m}$. Take z from near-identity plaquette lifts after removing the prescribed seam-centre factors. Keep the common denominator $\det H_1$: sector-dependent constants then remain in relative sector weights. Ordinary periodic

SU(2) backgrounds here have $m = 0$; $m = 1$ requires an explicitly twisted boundary problem, or an external centre-flux insertion. Summing magnetic sectors would be a further choice of theory.

For an explicit flat representative choose constant transitions $\Gamma_2\Gamma_3 = (-1)^m\Gamma_3\Gamma_2$. Their commuting adjoint rotations have joint phases (ϕ_2, ϕ_3) , and the eigenvalues are

$$L_m(r, s; \phi) = 4 - 2 \cos \frac{2\pi r + \phi_2}{n_2} - 2 \cos \frac{2\pi s + \phi_3}{n_3}, \quad P_m = (8 + L_m)^{-1}. \quad (51)$$

For $m = 0$, commuting $\Gamma_j = \exp(\theta_j T_3)$ give phases $(0, 0)$ and $\pm(\theta_2, \theta_3)$. For $m = 1$, take $\Gamma_2 = i\sigma_1$, $\Gamma_3 = i\sigma_2$ (Pauli matrices): conjugation fixes its own axis and reverses the other two, giving phase pairs $(0, \pi), (\pi, 0), (\pi, \pi)$. Formula (51) follows by translating a plane wave once around each cycle. Curved backgrounds use $C_{U,m}$ directly and retain their continuous Wilson-loop data within the sector. In §10's strip, H in $\cos(p + H)$ is the rotation **per link**; the free row cycle is $\exp(n_3 H T_3)$, with adjoint phase $n_3 H$ modulo 2π . Fixing two based cycles, or fixing $m = 0$, leaves those row holonomies curvature-dependent. Discrete flux labels alone therefore leave the transport term of (46) present.

With a temporal closure the electric-sector prescription is $Z_{e,m} = \frac{1}{4} \sum_{k \in \mathbb{Z}_2^2} (-1)^{e \cdot k} Z_{k,m}$, $e \in \mathbb{Z}_2^2$. Apply it to the assembled reference weights with their temporal seam data retained, before taking a logarithm. Electric flux is a quantum sector of the assembled kernel; a single classical background U specifies neither e nor a Fourier transform of log references. The adjoint local bounds survive either spatial twist; relative electric-sector estimates additionally require control of cancellation in this signed projection.

12.4 Classical verdict and consequence for STATE

Proposition 7 gives, for $\alpha = 1/2 - \delta > 2\delta$, $0 < \delta < 1/6$,

$$|F^{\text{cl}}/t - \mathcal{D}_{0,\text{cl}}^{\text{cov}}| \leq K t^\alpha \|z\|^2/t. \quad (52)$$

This proves classical weak $P(\alpha)$ on the stated equal-layer slice, uniformly in plane size; (48) also permits the full reference (47) in (52) after adding $Bt^{1-2\delta}N + \frac{9}{2}N3^{-n}$, admissible under the size clause. Classical coupling shifts may be zero. Identification with the normalized free energy remains **formal**. The Gaussian determinant bound settles only that frozen model's one-loop term; the actual saddle Hessian, Haar/heat-kernel amplitudes, normalized loop remainder, barrier moments, image/large-field comparison and general boundary layers remain. **Consequence for STATE:** retain this covariant reference and its sector weights; next prove the normalized covariant comparison, then stability under perturbed actions. Equivalence to the original fixed-frame $P(\alpha)$ on the full small-field set still requires the transport-value estimate, and electric-sector projection requires its own relative bound.

13. Round 15: the normalized covariant comparison

GPT-6 Astra, 2026-09-29; written bounds, unrefereed. Throughout this section the layers are equal, cut flux is zero, and all free-cycle holonomies remain in C_U . Put $\alpha = 1/2 - \delta$, $0 < \delta < 1/6$, $\varepsilon = t^\alpha$, and impose Proposition 7's smallness conditions.

After refereeing §13 (Claude, 2026-09-29). ACCEPT (53)–(58) as finite-plane written proofs under the stated chart and time conditions. Checked: (53) from the block- ℓ^1 contraction with $\|H_U^{-1}\|_{1 \rightarrow 1} \leq \frac{1}{2}$ and $\|\xi_0\|_{1,b} \leq \sum_p |z_p|$; (54) from rank 12 per face, $\sum_p \sum_{e \in \partial p} |\xi_{*e}| = 2\|\xi_*\|_{1,b}$ and $\frac{1}{2} \cdot \frac{1}{3} \cdot 12L_F(1 + \frac{16}{7}) = \frac{46}{7}L_F$; the moment bound (55) by integration by parts against $|y_e|^{k-2}y_e$, whose divergence in $\mathbb{R}^3$ is $(k+1)|y_e|^{k-2}$, with the barrier term nonnegative because $|\xi_{*e}| \leq 2\varepsilon/7 < \eta$, and diagonal dominance $(5 - \kappa_r) - (3 + \kappa_r) = 2 - 2\kappa_r \geq \frac{3}{2}$ at the maximizing edge; the cubic Taylor bound with 16 products per face; Šidák's rectangle, which stays in $|\xi_e| \leq 9\varepsilon/7 < \eta = 2\varepsilon$, with coordinate variances at most $t/3.75$; the assembly of C_0 from

(43), (54), (56) and $e^{-t^{-2\delta}/2} \leq B_\delta t^\alpha$; (57) with the counts 2 and 3 and $\frac{38}{7}$; and the $k = 2$ bound $M_2^2 \leq 3t + 3D_J t M_2$ with net dominance at least 1 when $\kappa_r, 9tD_J \leq \frac{1}{4}$. The condition $\kappa_r \leq \frac{1}{4}$ shrinks the chart to $c_0 \lesssim 1/(32L_F)$, which moves more of the integral into the large-field comparison that §13 leaves open. One remark for the iteration: the remainder $2A_L N \sqrt{t}$ in (56) is additive and depends on the whole field through the interpolated measure, including flat holonomies. The stated form of $P(\alpha)$ admits it. A multi-step argument will need it as a sum of quasi-local terms; the uniform moments (55) and Lemma 1's covariant decay are the ingredients for that rewriting.

13.1 Actual saddle determinant (Part 1)

Write $A_U = \nabla^2 E_U(\xi_*)$, $\Delta_U = A_U - H_U$ and $\|\xi\|_{1,b} = \sum_e |\xi_e|$. The contraction proof of Proposition 7 also works in block ℓ^1 : symmetry gives the column bounds from its row bounds, and $\|C_U^* z/2\|_{1,b} \leq 2 \sum_p |z_p|$. Hence

$$\|\xi_*\|_{1,b} \leq \frac{\sum_p |z_p|}{1 - K\varepsilon/2} \leq \frac{8}{7} \sum_p |z_p|. \quad (53)$$

Each face Hessian difference has rank at most 12 and operator norm at most $L_F(|z_p| + \sum_{e \in \partial p} |\xi_{*e}|)$, by §4's mean-value bound with the orthogonal incidence maps held fixed. The bridge Hessian is exactly $4I$. Thus the trace norm satisfies $\|\Delta_U\|_{\text{tr}} \leq 12L_F(\sum_p |z_p| + 2\|\xi_*\|_{1,b})$. Every $H_U + s\Delta_U \geq 3I$, $0 \leq s \leq 1$, so integrating $\frac{1}{2} \text{Tr}[(H_U + s\Delta_U)^{-1} \Delta_U]$ proves

$$\left| \frac{1}{2} \log \frac{\det A_U}{\det A_1} - d_U \right| \leq C_D \sum_p |z_p|, \quad C_D = \frac{46}{7} L_F. \quad (54)$$

Here $A_1 = H_1$ exactly. More generally $z = 0$ gives $\xi_* = 0$ and $A_U = H_U$, even with nontrivial flat holonomies. This is a local field-dependent bound; the trace norm replaces any need to discard transports in the resolvent. Both (53) and (54) are uniform in N . For $t \leq 1$ and $\alpha \leq 1/2$, $|z_p| \leq (t^\alpha + |z_p|^2/t^\alpha)/2 \leq t^\alpha(1 + |z_p|^2/t)/2$, giving weak $P(\alpha)$ with constant $C_D/2$. The coarser bound $6NK\varepsilon/4$ also follows from $\|\Delta_U\| \leq K\varepsilon \leq 1/4$ and $|\log(1+x)| \leq |x|/(1-|x|)$; it is admissible in the additive budget, while (54) additionally vanishes at zero curvature.

13.2 Laplace remainder and the soft barrier (Part 2)

Choose $2\varepsilon < c_0 \leq 1/16$ still smaller if necessary so that $\kappa_r = 8L_F(\varepsilon + 4c_0) \leq 1/4$. Define $y = \xi - \xi_*$, $S_2 = (E_U(\xi_*) + y^T A_U y/2)/t$, $R = E_U/t - S_2$, and $V_\lambda = S_2 + \lambda R + W_\eta$, $0 \leq \lambda \leq 1$. The partition function $Z_E = \int e^{-E_U/t - W_\eta}$ equals $e^{\|z\|^2/(4t)} Z_s^0$, where Z_s^0 is (2) with J removed. Thus its normalization includes precisely the classical old-face ratio. On the whole chart, the unbarriered Hessians have diagonal blocks $\geq (5 - \kappa_r)I/t$, off-diagonal block majorants summing to $(3 + \kappa_r)/t$, and lower operator bound $(4 - \kappa_r)I/t$. Indeed H_U has diagonal $5I$ and six off-diagonal incidences of norm $1/2$; the eight local Hessian blocks per row give the stated error majorant. Convex interpolation preserves these bounds. The fixed convex barrier preserves the lower bound, hence (13), along the entire path.

Here is the additional saddle-centering argument. Proposition 7 gives $|\xi_*|_\infty \leq 2\varepsilon/7 < \eta$, so $y_e \cdot \nabla w_\eta(\xi_e) \geq 0$. Integration by parts against $|y_e|^{k-2} y_e$, followed by Hölder at an edge maximizing $M_k = (\langle |y_e|^k \rangle_\lambda)^{1/k}$, gives $(2 - 2\kappa_r)M_k^k \leq t(k+1)M_k^{k-2}$. Thus, for $k \geq 2$,

$$\sup_{\lambda, e} \langle |y_e|^k \rangle_\lambda \leq M_k^{\text{bd}} t^{k/2}, \quad M_k^{\text{bd}} = [2(k+1)/3]^{k/2}. \quad (55)$$

The barrier's boundary decay in (2) justifies these integrations; regularizing $|y_e|^{k-2} y_e$ at zero gives the same inequality. In particular the probability of $|\xi_e| \geq r > \varepsilon$ is at most $M_k^{\text{bd}} t^{k/2}/(r-\varepsilon)^k$,

including layers near c_0 . This controls ordinary local moments; expectations of the unbounded barrier Hessian require separate hypotheses, and are unused here.

Taylor's integral remainder gives $|R| \leq L_F \sum_p (\sum_{e \in \partial p} |y_e|) (\sum_{e \in \partial p} |y_e|^2) / (6t)$. There are 16 products per face, each bounded in expectation by $M_3^{\text{bd}} t^{3/2}$ from (55), with $M_3^{\text{bd}} = (8/3)^{3/2}$. Put $A_L = (8/3)L_F(8/3)^{3/2}$. Integrating $\partial_\lambda \log Z_\lambda = -\langle R \rangle_\lambda$ yields $|\log Z_E - \log Z_{2,W}| \leq A_L N \sqrt{t}$. Expansion about the actual Hessian absorbs the ε -quadratic term into (54); the remaining Taylor error starts cubically.

Let $G_U = \int_{\mathbb{R}^{6N}} e^{-S_2}$ and $q = 2e^{-\varepsilon^2/(2t)} \leq 1/2$. The centered Gaussian has scalar variances at most $t/3$. The rectangle $|y_{e,a}| \leq \varepsilon/\sqrt{3}$ lies inside $W_\eta = 0$. Šidák's Gaussian rectangle inequality gives $Z_{2,W}/G_U \geq (1-q)^{6N}$, while $Z_{2,W}/G_U \leq 1$. Input: Šidák 1967, Corollary 1, p. 628 (**passage**, original paper; Crossref metadata verified 2026-09-29). Consequently the identity-normalized remainder obeys

$$\left| \log \frac{Z_E(U)}{G_U} - \log \frac{Z_E(1)}{G_1} \right| \leq 2A_L N \sqrt{t} + 48N e^{-t^{-2\delta}/2}. \quad (56)$$

This includes the barrier cost for every barrier satisfying (2), regardless of its growth near c_0 . Put $B_\delta = (\alpha/(\delta e))^{\alpha/(2\delta)}$; maximizing $x^{\alpha/(2\delta)} e^{-x/2}$ gives $e^{-t^{-2\delta}/2} \leq B_\delta t^\alpha$. Combining (43), (54) and (56) proves the normalized, amplitude-free bound $|\log[Z_E(U)/Z_E(1)] - \mathcal{D}_0^{\text{cov}}(U)| \leq C_0 t^\alpha \sum_p (1 + |z_p|^2/t)$, with $C_0 = K + C_D/2 + 2A_L + 48B_\delta$. The normalized field dependence is controlled by the allowed additive term; (56) permits a flat-holonomy remainder as well as curvature.

13.3 Haar/heat-kernel amplitudes (Part 3)

Include the external old-face amplitude once: write $\mathcal{J} = J + B_{\text{amp}}$ as one local edge term, one mid-face term and one old-face term per cell, with background-independent constants removed. Define $D_J \geq 1$ as the maximum of their first block derivatives (in z, ξ_i) and second ξ -block derivatives on §4's compact charts, uniformly for $0 < t \leq 1/2$. Section 4's analytic amplitude/image argument makes this supremum finite. At $z = \xi = 0$ all local values equal their identity-background values, regardless of the transports. Incidence counting and (53) give

$$|\mathcal{J}(U, \xi_*) - \mathcal{J}(1, 0)| \leq D_J (2 \sum_p |z_p| + 3 \|\xi_*\|_{1,b}) \leq \frac{38D_J}{7} \sum_p |z_p|. \quad (57)$$

For interpolation by $s\mathcal{J}$, impose also $9tD_J \leq 1/4$. Its Hessian block row majorant is $9D_J$, and its saddle score is at most $3D_J$ per edge. The proof of (55), now with $k = 2$, gives $M_2^2 \leq 3t + 3D_J t M_2$, hence $M_2 \leq (\sqrt{3} + 3D_J)\sqrt{t}$ for $t \leq 1$. Thus the interpolated potential stays convex, and $|\langle \mathcal{J} - \mathcal{J}(U, \xi_*) \rangle_s| \leq 6D_J(\sqrt{3} + 3D_J)N\sqrt{t}$. Integrating in s , identity-normalizing, and using (57) proves, with $Z_E^J = \int e^{-E_U/t - W_\eta - \mathcal{J}}$,

$$\left| -\log \frac{Z_E^J(U)}{Z_E^J(1)} - \mathcal{D}_0^{\text{cov}}(U) \right| \leq C t^\alpha \sum_p (1 + |z_p|^2/t), \quad C = C_0 + \frac{19D_J}{7} + 12D_J(\sqrt{3} + 3D_J). \quad (58)$$

Consequence for STATE

Equation (58) proves normalized weak $P(1/2 - \delta)$ **against (47)** for one barriered step on the equal-layer, zero-cut-flux slice, uniformly in plane size, with free-cycle holonomies retained and explicit chart smallness conditions. The bound allows zero additional coupling shifts. The next cell-2 task is general boundary layers; comparison to the full integral outside this chart (images/large fields), fixed-frame equivalence, electric projection, iteration and stability under perturbed actions remain. The present amplitude bound already includes images inside the compact chart.

14. Round 16: unequal layers and a quasi-local remainder

GPT-6 Astra, 2026-09-29; written bounds, unrefereed. The cut-face identity is $P_e Q_e = U_{f_e}$ (series/parallel, Proposition 1, **passage**). Consequently zero cut flux means $P_e = Q_e^{-1}$: after transporting both layers to the mid-vertex frames, they coincide. Unequal layers in those frames require cut flux. We treat $|X_e| \leq \varepsilon$ below, and recover the requested zero-cut-flux case by setting $X = 0$.

14.1 Unequal layers: reference and comparison (Part 1)

Write the endpoints as $m_{*e} \exp(\pm b_e/2)$, $|b_e| = |X_e|$, and put $Y_p = \log m_{* \partial p}$, $w = (x + y)/2$, $a = Y - w$. Here x, y are the two transverse face logs in their common basepoint frames; assume $|x_p|, |y_p|, |b_e| \leq \varepsilon$. Keep all midpoint cycles free. With $C = C_{m*}$, $P = (8 + CC^*)^{-1}$, $H = 4 + C^*C/2$, the covariantization of (29), including its determinant, is

$$\begin{aligned} E_w^{(2)}(\xi) &= 2\|\xi\|^2 + \frac{1}{4}\|w + C\xi\|^2 - \frac{1}{8}(\|x\|^2 + \|y\|^2), \\ \mathcal{D}_{xy}^{\text{cov}} &= \frac{1}{2t}\langle x + y, P(x + y) \rangle - \frac{1}{8t}(\|x\|^2 + \|y\|^2) + \frac{1}{2} \log(\det H / \det H_1). \end{aligned} \quad (59)$$

The actual energy is $E = \sum_e B(b_e, \xi_e) + \sum_p F(Y_p, O_{pe} \xi_e) - (\|x\|^2 + \|y\|^2)/8$, with the four signed rotations of §4. Define β as half the maximum of the suprema of $\|D_b^2 D_\xi^2 B\|, \|D_b^2 D_\xi^3 B\|$ on §4's compact chart. Evenness in b and $B(0, \xi) = 2|\xi|^2$ give $\|D_\xi^2 B - 4I\|, \|D_\xi^3 B\| \leq \beta|b|^2$ there. At zero the exact Hessian is $4\Pi_b + 4h_b\Pi_b^\perp$, $h_b = (|b|/4) \cot(|b|/4)$, with the continuous value at $b = 0$; the bridge gradient vanishes. Thus $\nabla E(0) = C^*Y/2$ exactly.

Let M be the supremum of the second derivative of the local face-log map in its four insertion blocks, using their sum norm, on the same chart; set $c_M = M/2$. Taylor expansion of the two endpoint products at insertion $\pm b/2$ cancels the linear terms and gives $|a_p| \leq c_M \sum_{e \in \partial p} |b_e|^2$. Consequently, writing $S = \|x\|^2 + \|y\|^2$ and $T = \sum_e |b_e|^2$, $\|a\|^2 \leq 8c_M^2 \varepsilon^2 T$ and $\|Y\|^2 \leq S + 16c_M^2 \varepsilon^2 T$; also $|Y_p| \leq 3\varepsilon$.

Put $K_u = \beta + 56L_F + 1$ and impose $3\varepsilon \leq 1/16$, $K_u \varepsilon \leq 1/4$, $2\varepsilon < c_0 \leq 1/16$ and $\kappa_u = \beta \varepsilon^2 + 8L_F(3\varepsilon + 4c_0) \leq 1/4$. The proof of (43), with Y replacing z , has Hessian row error $\leq K_u \varepsilon$ on the radius- ε ball and initial iterate $\|\xi_0\|_\infty \leq 3\varepsilon/4$. Its contraction gives $\|\xi_*\|_\infty \leq 6\varepsilon/7$, $\|\xi_*\|_{1,b} \leq (8/7) \sum_p |Y_p|$ and $\|\xi_*\|_2 \leq \sqrt{2}\|Y\|_2/3$. The classical comparison is

$$|\min E - \min E_w^{(2)}| \leq C_{\text{cl}} \varepsilon (S + T), \quad C_{\text{cl}} = K_u(1 + 16c_M^2) + c_M + 2c_M^2. \quad (60)$$

Indeed comparison first to $E_Y^{(2)}$ costs $K_u \varepsilon \|Y\|^2$; the remaining difference is $2\langle Y, PY \rangle - 2\langle w, Pw \rangle$, bounded by $\|w\|\|a\|/2 + \|a\|^2/4$ since $\|P\| \leq 1/8$. For $A = \nabla^2 E(\xi_*)$, (54)'s trace proof now gives

$$\left| \frac{1}{2} \log \frac{\det A}{\det A_1} - \frac{1}{2} \log \frac{\det H}{\det H_1} \right| \leq C_D \sum_p |Y_p| + \frac{\beta}{2} T, \quad C_D = 46L_F/7. \quad (61)$$

The bridge contributes trace norm $3\beta T$; all interpolating Hessians exceed $3I$. The moments (55) survive with κ_u . The Gaussian rectangle still lies in the barrier-free region because $6\varepsilon/7 + \varepsilon < 2\varepsilon$. Equation (56) holds with A_L replaced by $A_u = A_L + (\beta/3)(8/3)^{3/2}$, accounting for the cubic bridge remainder on $2N$ edges. Enlarge $D_J \geq 1$ to include the first derivatives in b, Y, x, y and the same second insertion derivatives of the local amplitudes. Then (57)'s right side becomes $D_J[\sum_e |b_e| + \sum_p (|x_p| + |y_p|) + (31/7) \sum_p |Y_p|]$. With $9tD_J \leq 1/4$ the centered amplitude cost remains $12D_J(\sqrt{3} + 3D_J)N\sqrt{t}$ after identity normalization.

Thus, for $\alpha = 1/2 - \delta$, $0 < \delta < 1/6$, $t \leq 1/2$ and $2e^{-t^{-2\delta}/2} \leq 1/2$, the full barriered normalized bound is

$$\left| -\log \frac{Z_E^J(U)}{Z_E^J(1)} - \mathcal{D}_{xy}^{\text{cov}} \right| \leq C_u t^\alpha \left[\sum_p \left(1 + \frac{|x_p|^2 + |y_p|^2}{t} \right) + \sum_e \left(1 + \frac{|X_e|^2}{t} \right) \right], \quad (62)$$

where an explicit loose choice is $C_u = 1000(1 + C_{cl} + C_D + \beta + c_M + c_M^2 + A_u + B_\delta + D_J + D_J^2)(1 + c_M)$. Use $\sum |Y| \leq \frac{1}{2} \sum (|x| + |y|) + 2c_M T$ and §13's scalar Young bound to assemble it. This is weak $P(\alpha)$ including the cut faces, with zero extra coupling shifts. Setting $X = 0$ gives $x = y = Y$ and (58) (the extra $2N$ is absorbed in its constant). For genuinely unequal layers, the transverse-only target additionally needs control of the explicit midpoint term $[2\langle Y, PY \rangle - 2\langle w, Pw \rangle]/t$ and the cut contributions in (60)–(61) by that smaller budget.

14.2 Local densities and the remaining decay hypothesis (Part 2)

Exact decomposition; spatial response conditional. Taylor-expand each face and bridge term at the actual saddle, assigning half of each edge remainder to each adjacent face. This gives $R = \sum_p R_p$ exactly. With $\mu_{\lambda,U}$ from §13.2 define

$$\ell_p(U) = - \int_0^1 [\langle R_p \rangle_{\lambda,U} - \langle R_p \rangle_{\lambda,1}] d\lambda, \quad |\ell_p(U)| \leq 2A_u \sqrt{t} \leq 2A_u t^\alpha. \quad (63)$$

Equation (55) proves this face by face with the same 16 cubic products as before; half-edge assignment adds $\beta(8/3)^{3/2}/3$ to the constant. Consequently $\sum_p \ell_p$ is precisely the identity-normalized $\log(Z_E/Z_{2,W})$. Assign amplitudes to faces in the same way and put $Q_p = \mathcal{J}_p(\xi) - \mathcal{J}_p(\xi_*)$. Integrating their expectations along the $s\mathcal{J}$ interpolation gives densities a_p for the centered amplitude log-ratio, with $|a_p| \leq 12D_J(\sqrt{3}+3D_J)\sqrt{t}$. These bounds imply the requested value bound with $1 + |z_p|^2/t$ on the equal-layer slice, even at $z_p = 0$. Their dependence on U still comes through the full measure and saddle.

Here is the precise decay supplied by Theorem 5 **if its soft-barrier realization condition holds uniformly along these interpolations**. For a potential perturbation θk_T supported in T , hold the local Taylor coefficients and saddle fixed, and retain the convexity bounds and, for R , (55)'s fourth moments throughout the perturbation. Set $b_T = \sup \|\nabla k_T\|_{L^2}$ and $d = d(\partial p, T)$. The derivative Taylor bound and (55) with $k = 4$ give $\|\nabla R_p\|_{L^2} \leq A_R = (320/3)(L_F + \beta)$; incidence counting gives $\|\nabla Q_p\|_{L^2} \leq A_J = 6D_J$. Using $\kappa \leq 1/2$ (including amplitudes), (21) proves

$$|\partial_\theta \langle f_p \rangle| = |\text{Cov}(f_p, k_T)| \leq \frac{2t}{7} A_f b_T e^{-\gamma d}, \quad \gamma = \log(12/5), \quad (f, A_f) = (R, A_R) \text{ or } (Q, A_J). \quad (64)$$

The uniformity in λ, s permits integration; (25) sums remote scores. For actual boundary changes the exact response also contains $\langle \partial_\theta f_p \rangle$ and the score $\partial_\theta V$. Their quasi-local bounds require compatible boundary coordinates: $\partial_\theta \xi_* = -A^{-1} \partial_\theta \nabla E(\xi_*)$ has Lemma 1 decay for a local link score, whereas independent face-log variations require the additional transport control identified in §7.4. Finally (56) also contains $b(U) - b(1)$, where $b(U) = \log(Z_{2,W}/G_U)$. Its rectangle bound controls only the total cost. Completing the requested quasi-local remainder requires a uniform soft-barrier realization of (22), the boundary-score bounds just specified, and local densities for this Gaussian barrier cost with summable decay and size $O(t^\alpha)$. Equations (63)–(64) establish the local small densities and the conditional response estimate; the full quasi-local assertion retains these premises.